Distributed and Embedded Multi-Agent Systems

Prof. D.Sc. Carlos Eduardo Pantoja[1]

1. Federal University of Science and Innovation (UFCI/RJ)

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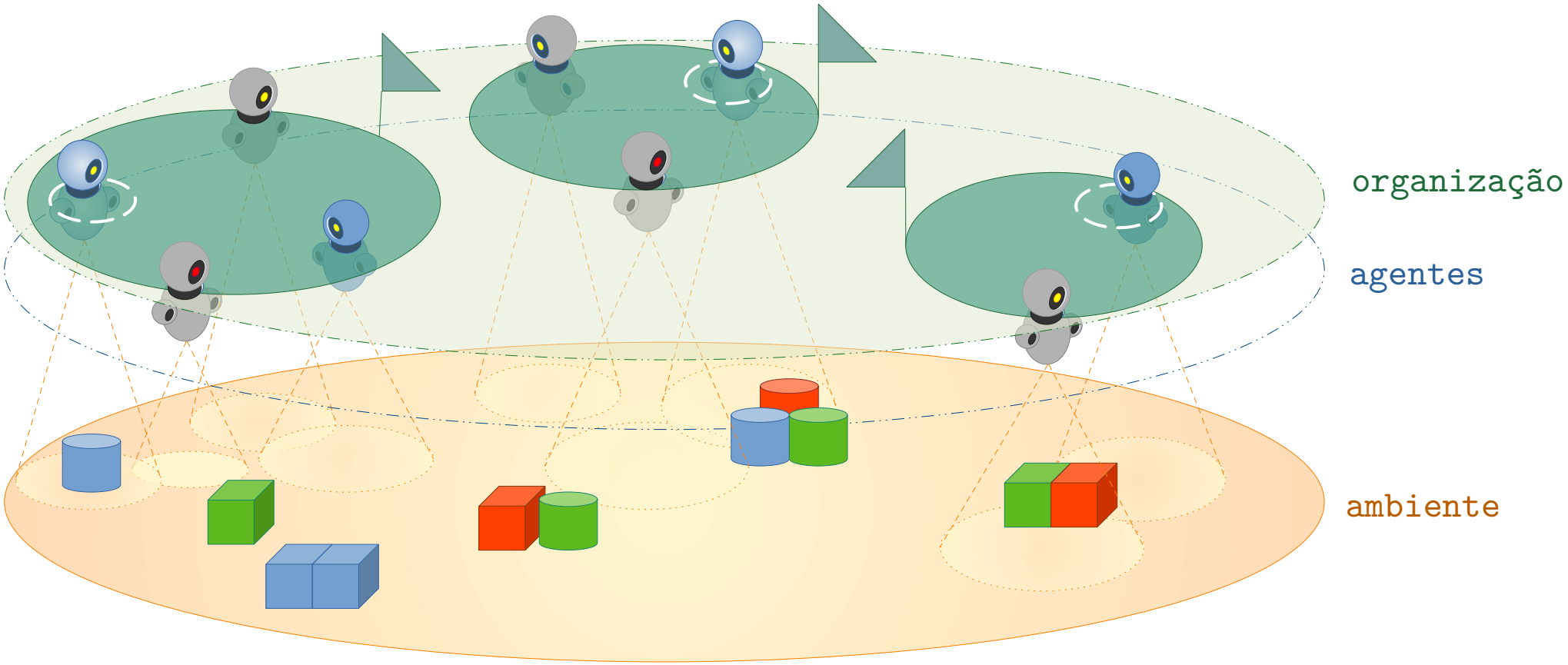
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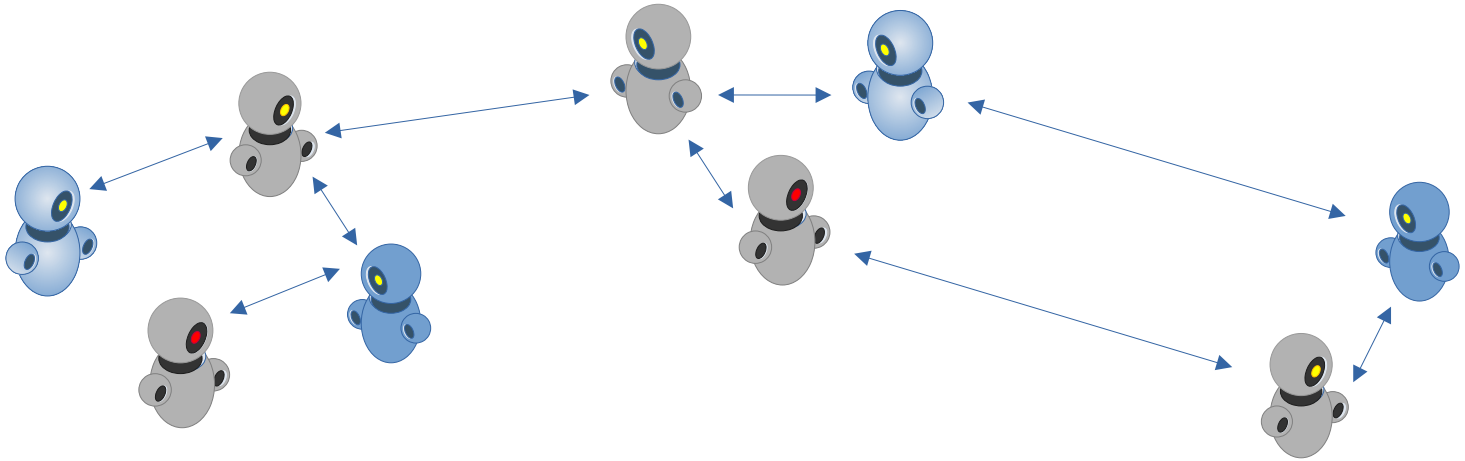
10th November, 2026

Multi-Agent Systems (MAS)



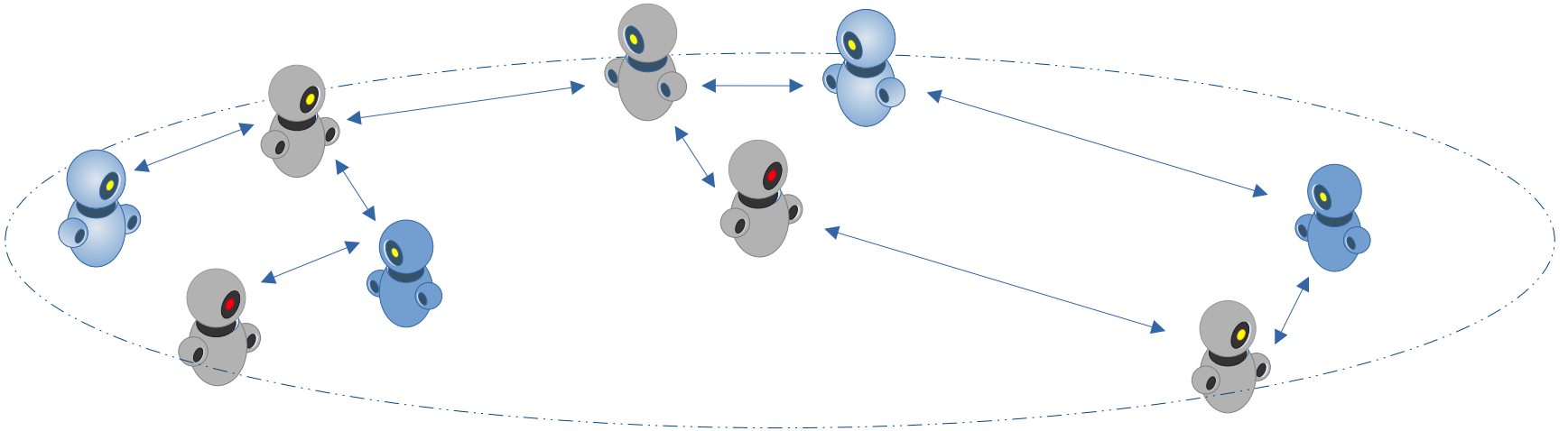
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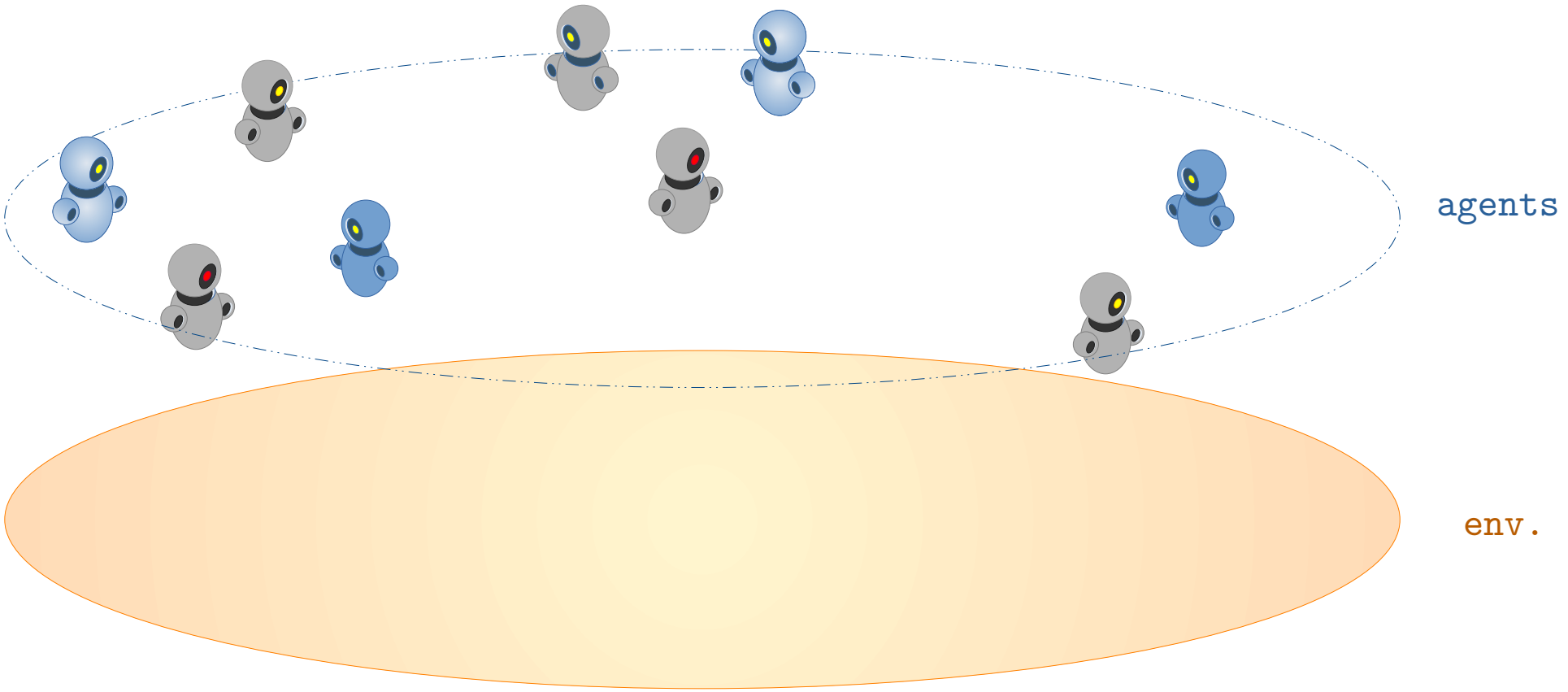
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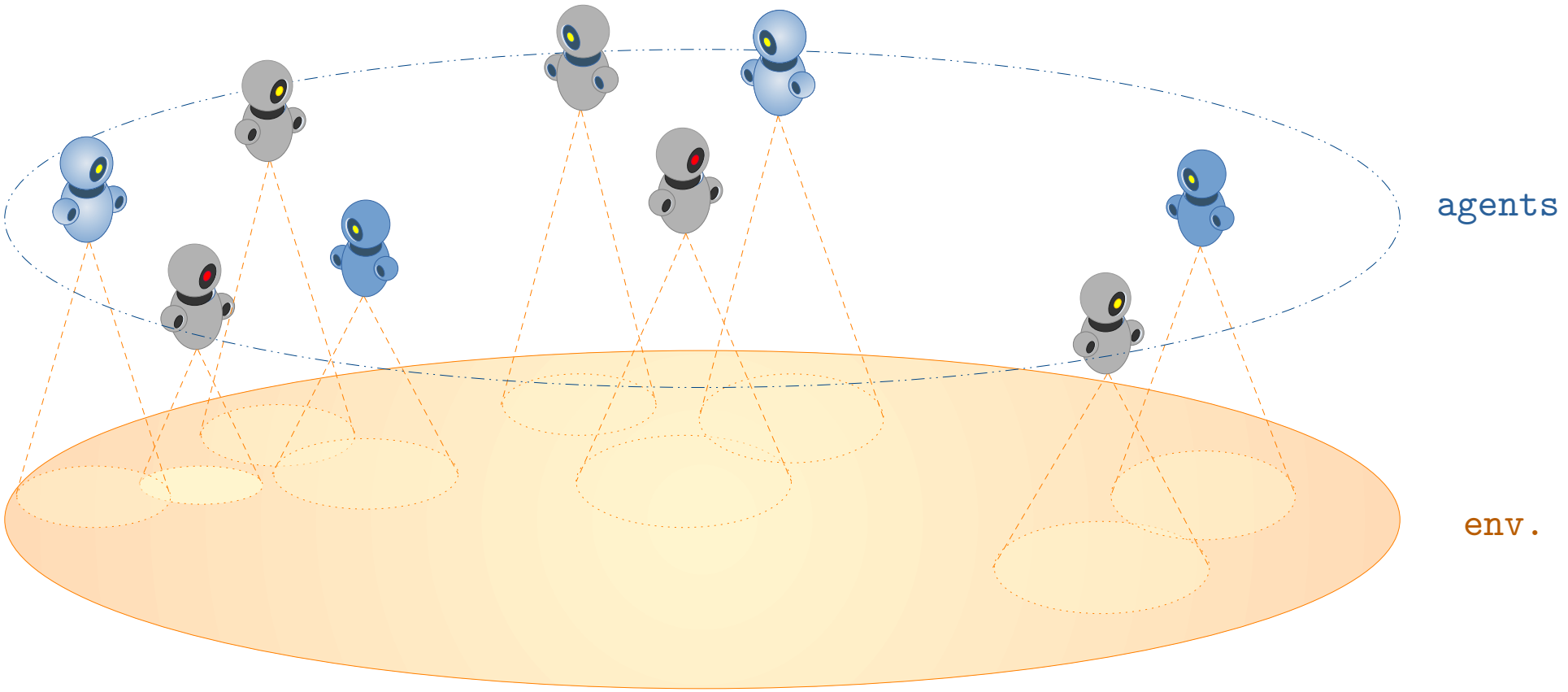
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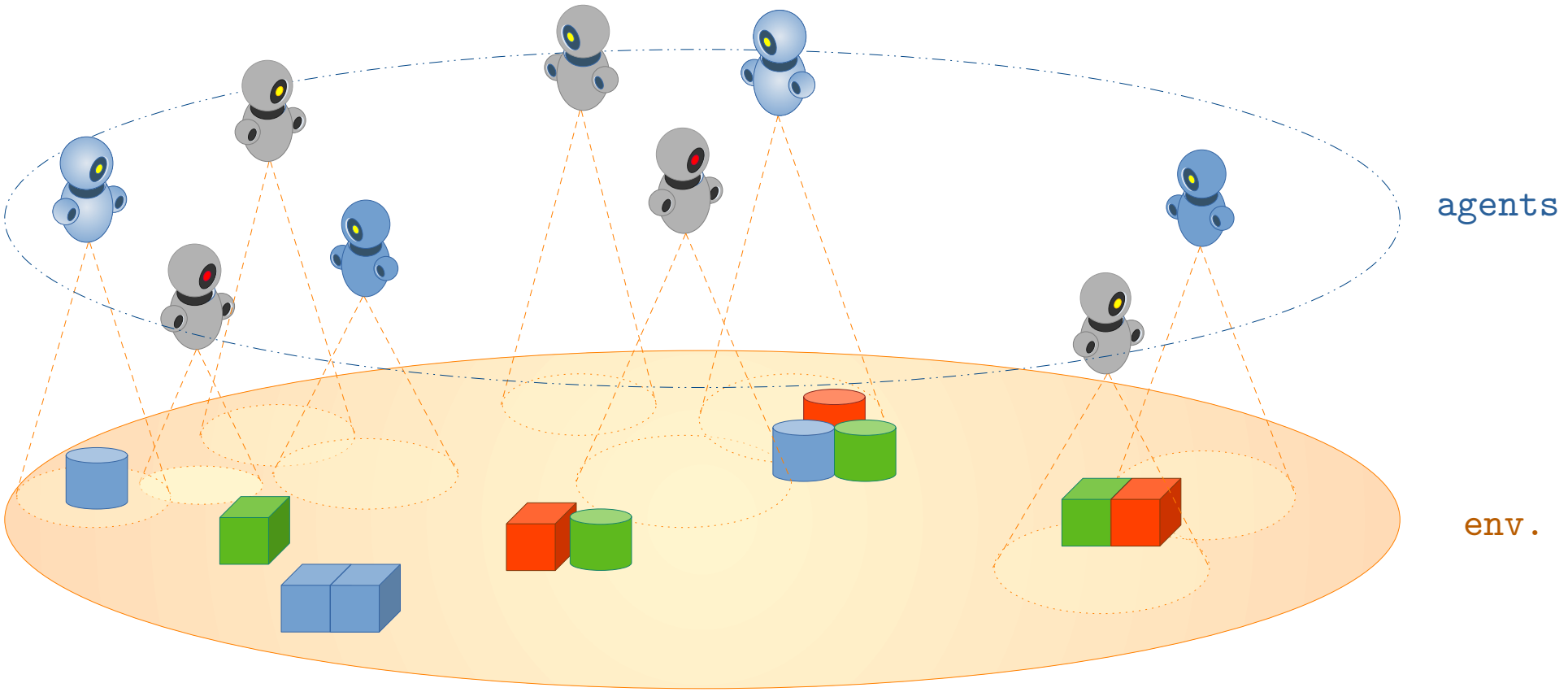
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Introduction

- Embedded Multi-Agent Systems
  - A system based on agents embedded in a device responsible for the device's autonomy, proactivity, and communication, controlling and accessing actuators, sensors, and communication infrastructures.

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Introduction

- Embedded Multi-Agent Systems
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  - **There is no remote control or external processing.**

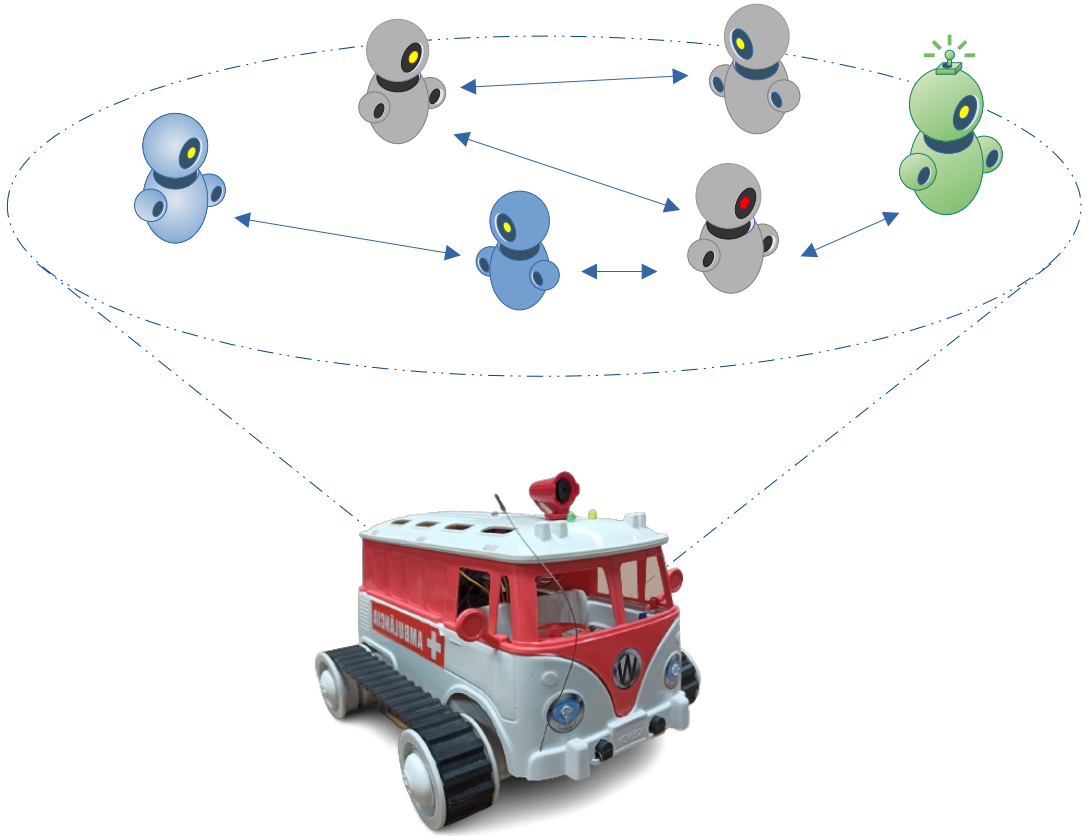
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Embedded MAS



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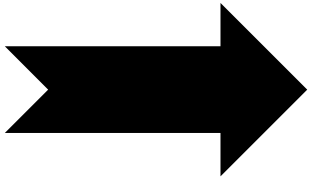
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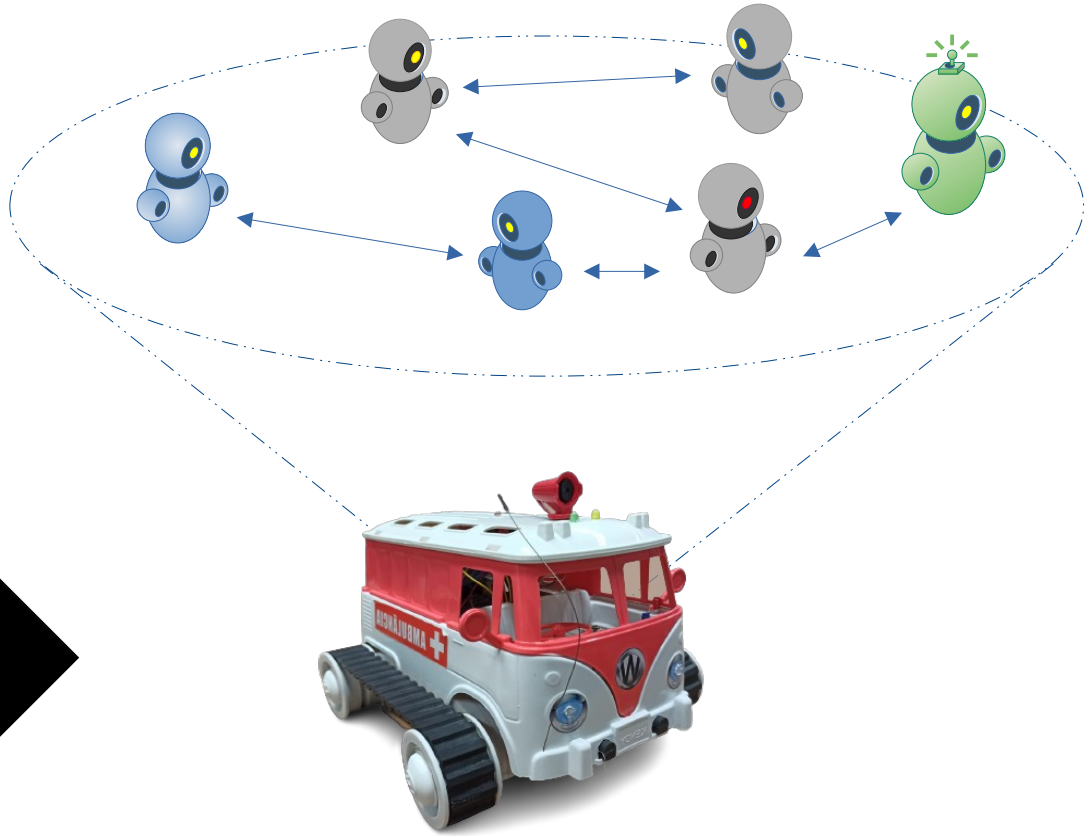
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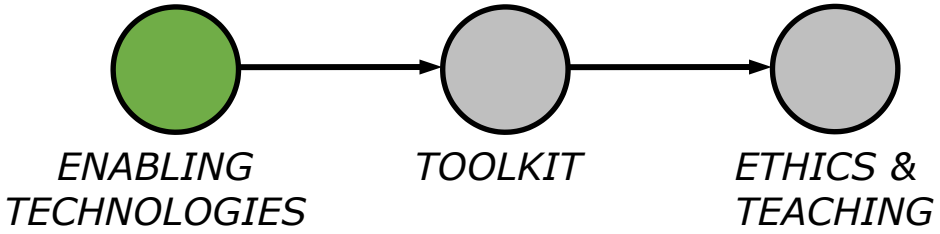
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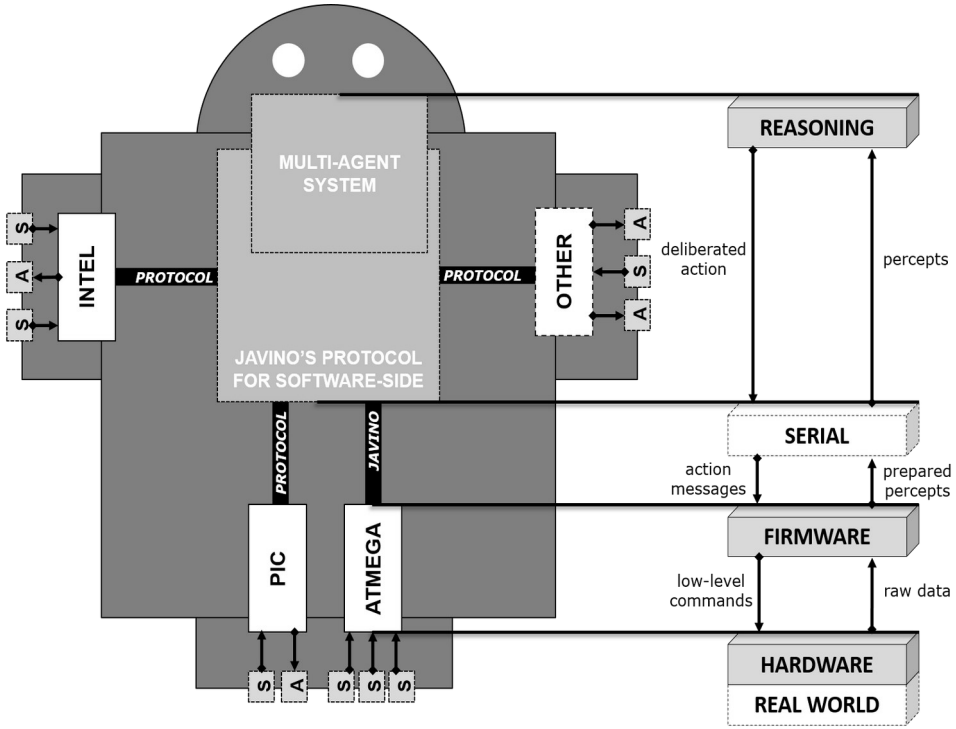


The Road So Far



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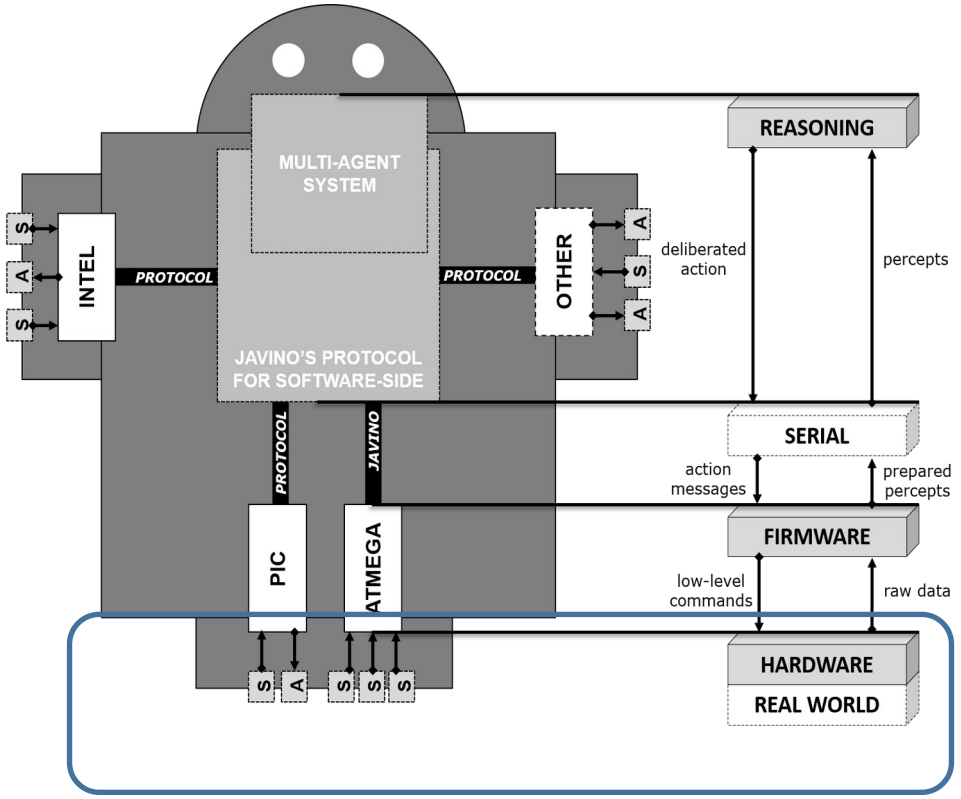
- Layers of an Embedded MAS



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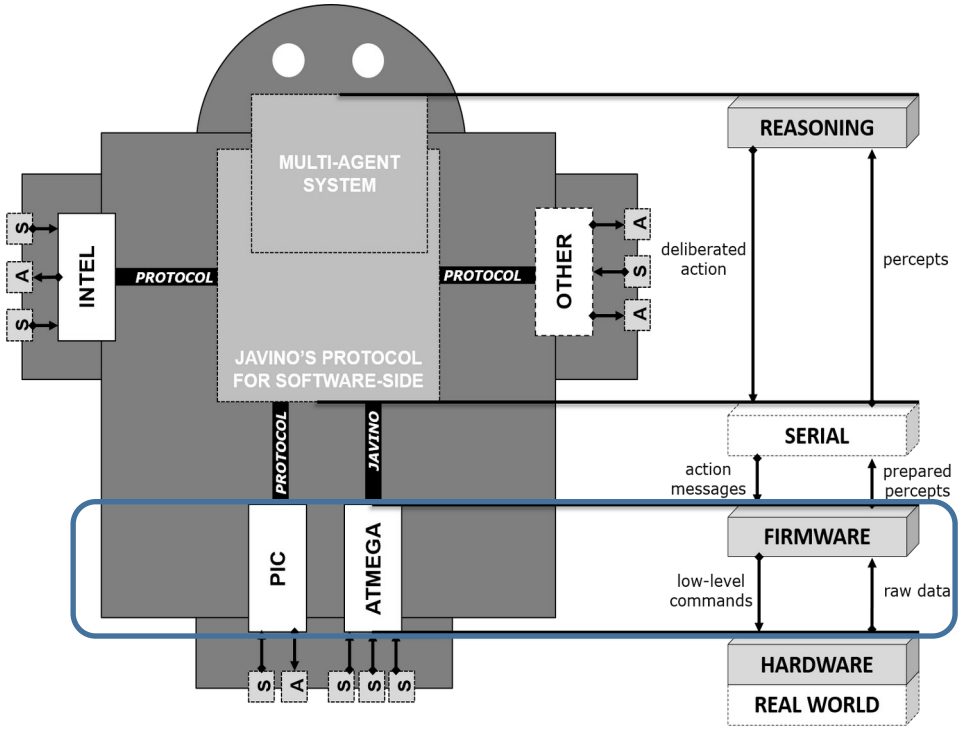
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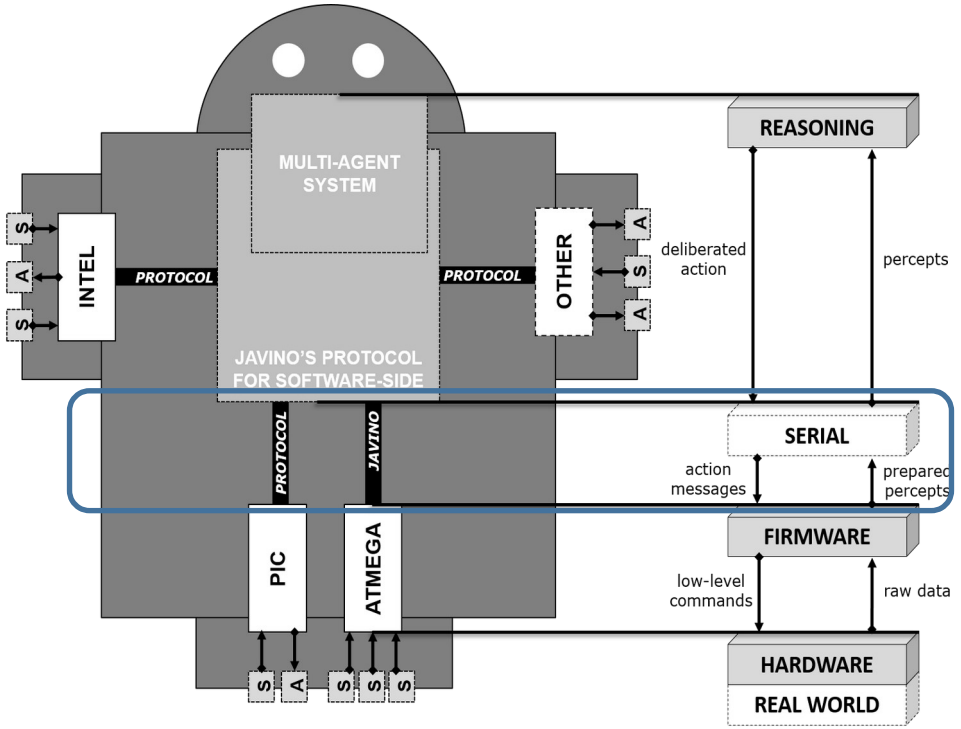
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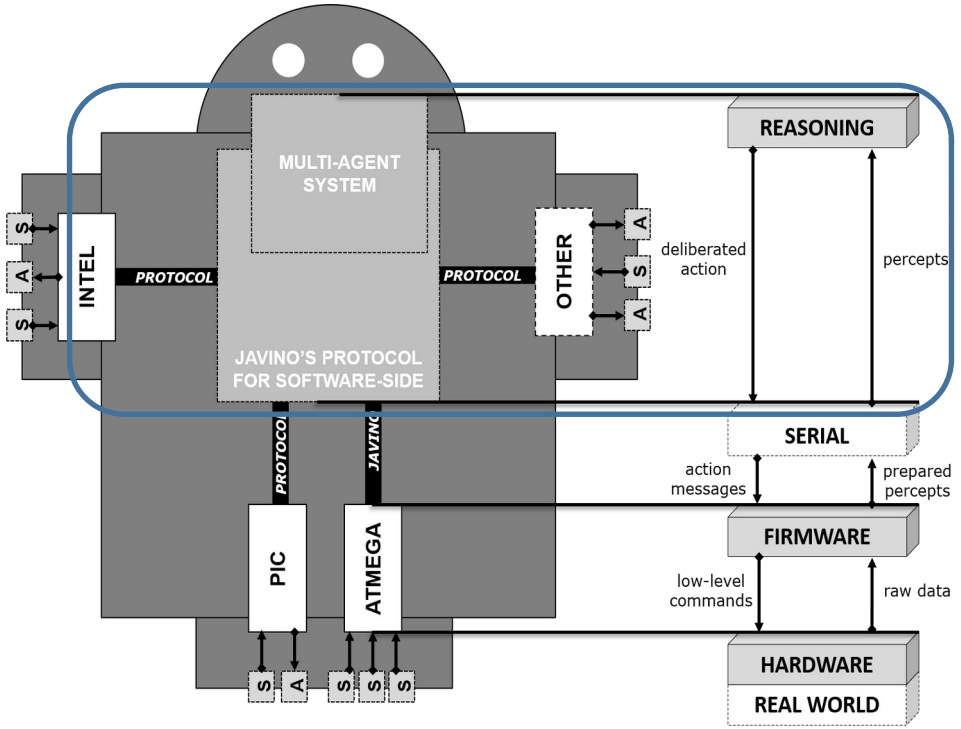
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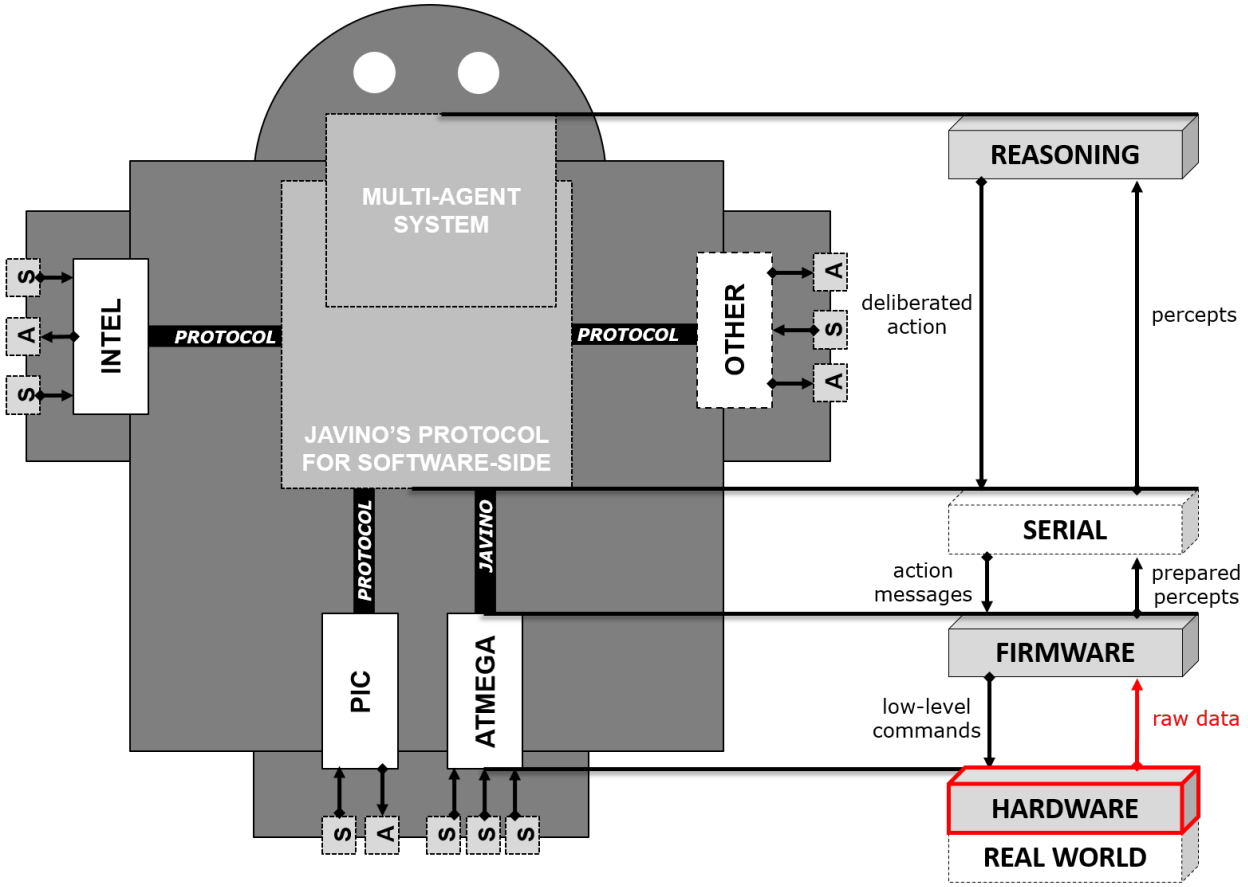
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  - **reasoning**: where the MAS is hosted.



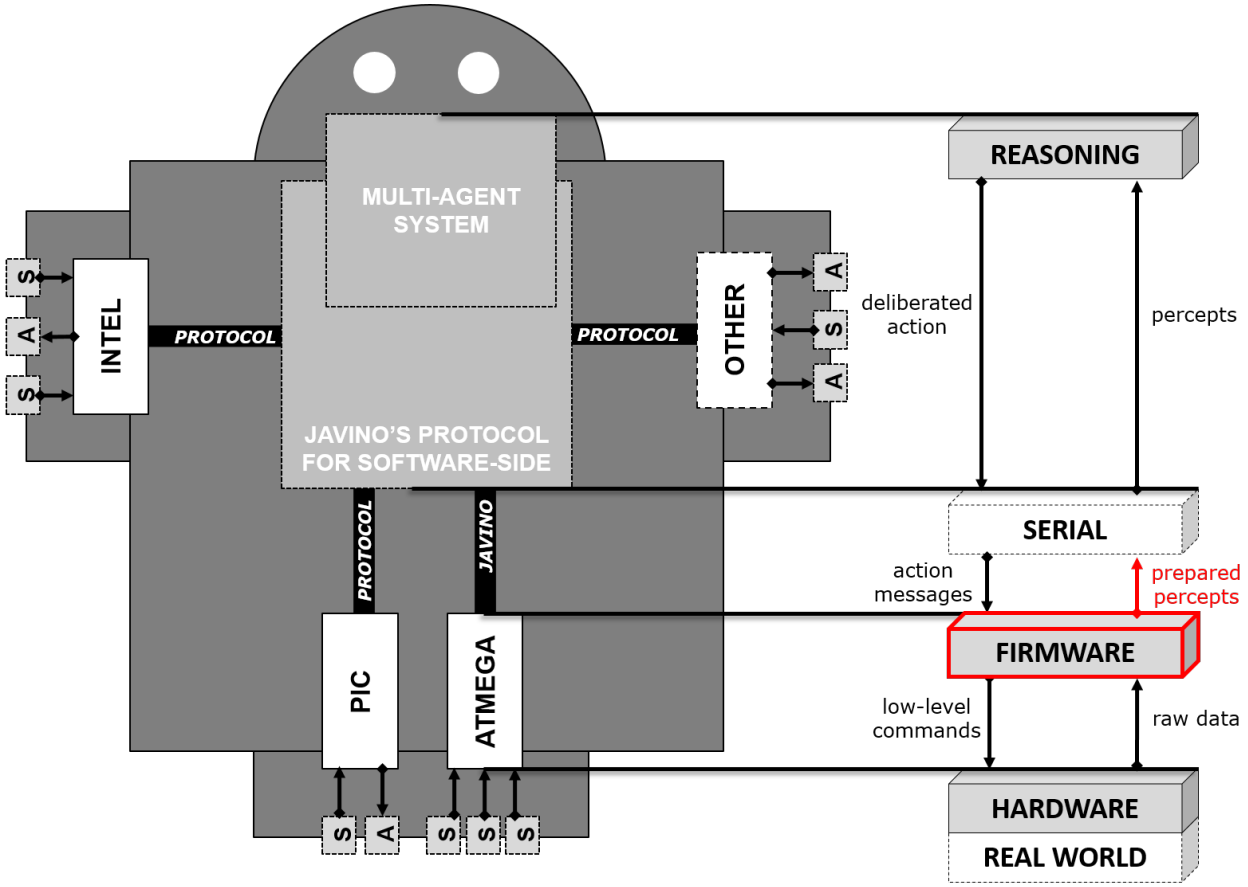
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Physical and Logical Architecture



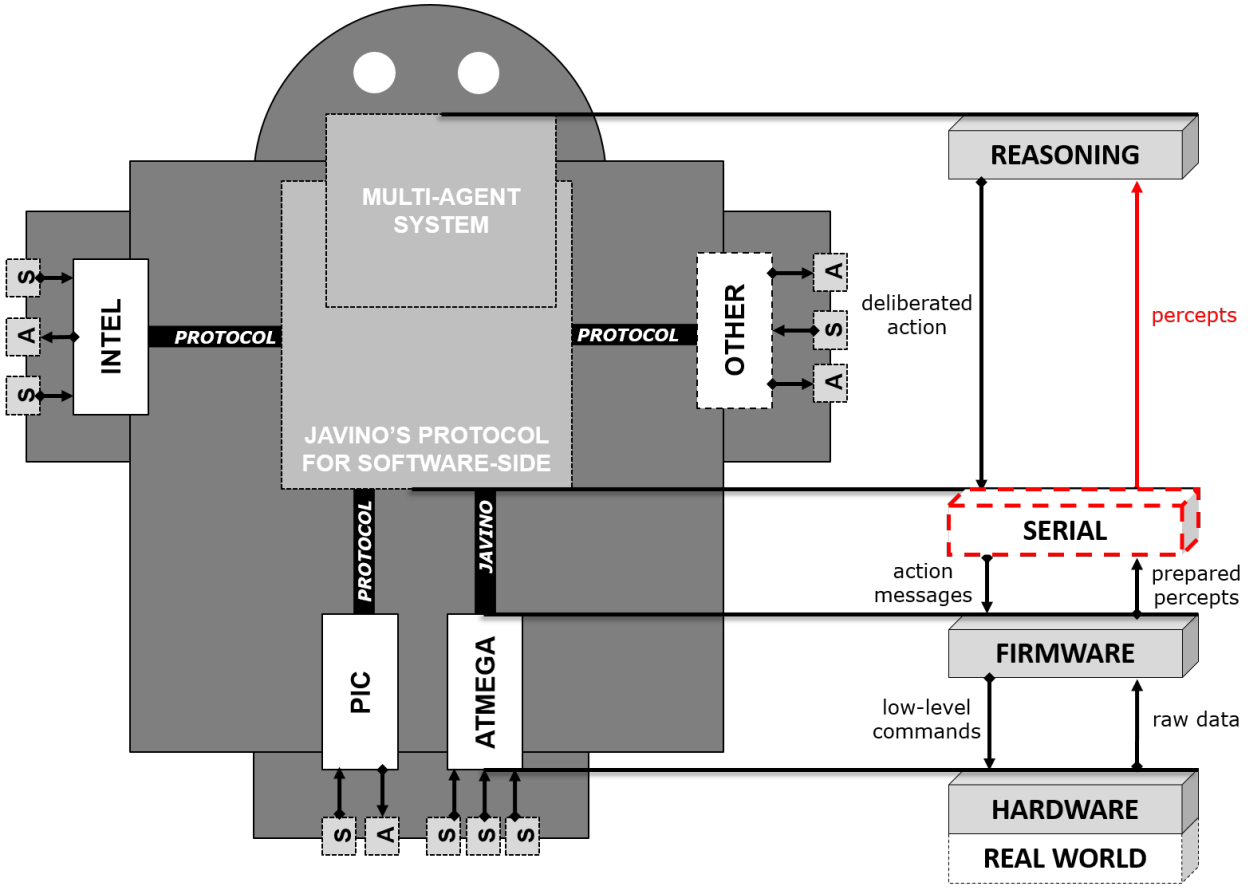
Sensors capture raw data from the real world and send it to one of the microcontrollers chosen for the project..

Physical and Logical Architecture



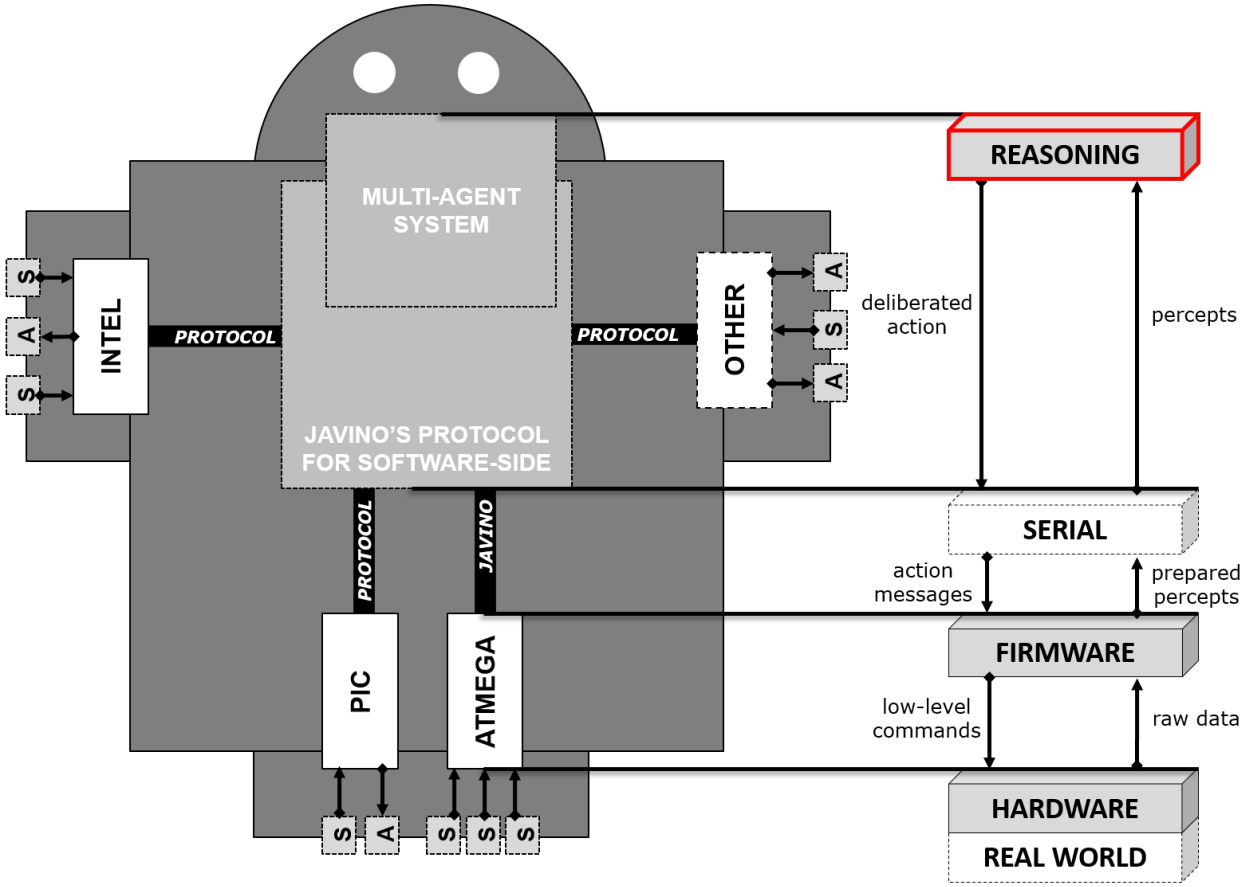
In microcontroller programming, raw data is transformed into insights based on the chosen agent-oriented programming language.

Physical and Logical Architecture



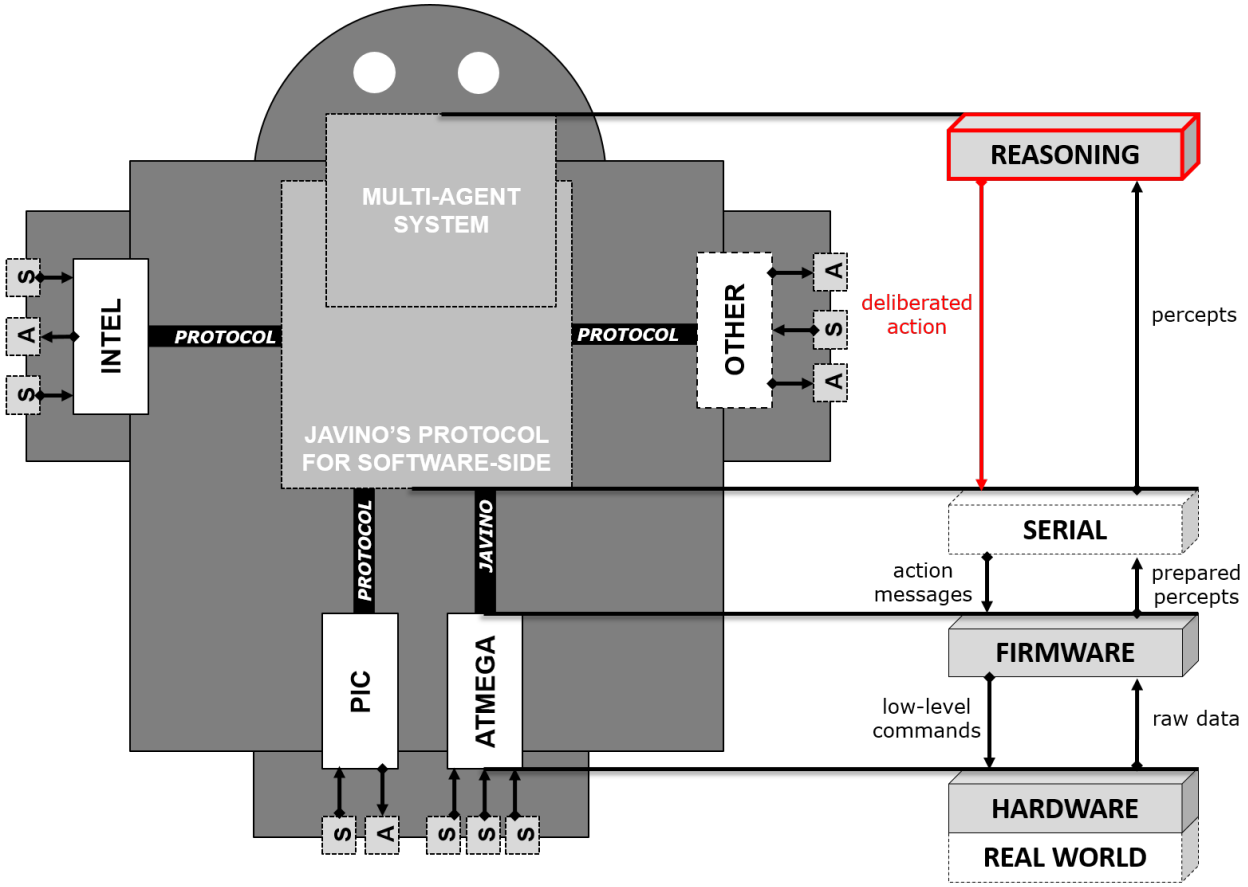
It sends the perceptions to the reasoning layer using serial communication.

Physical and Logical Architecture



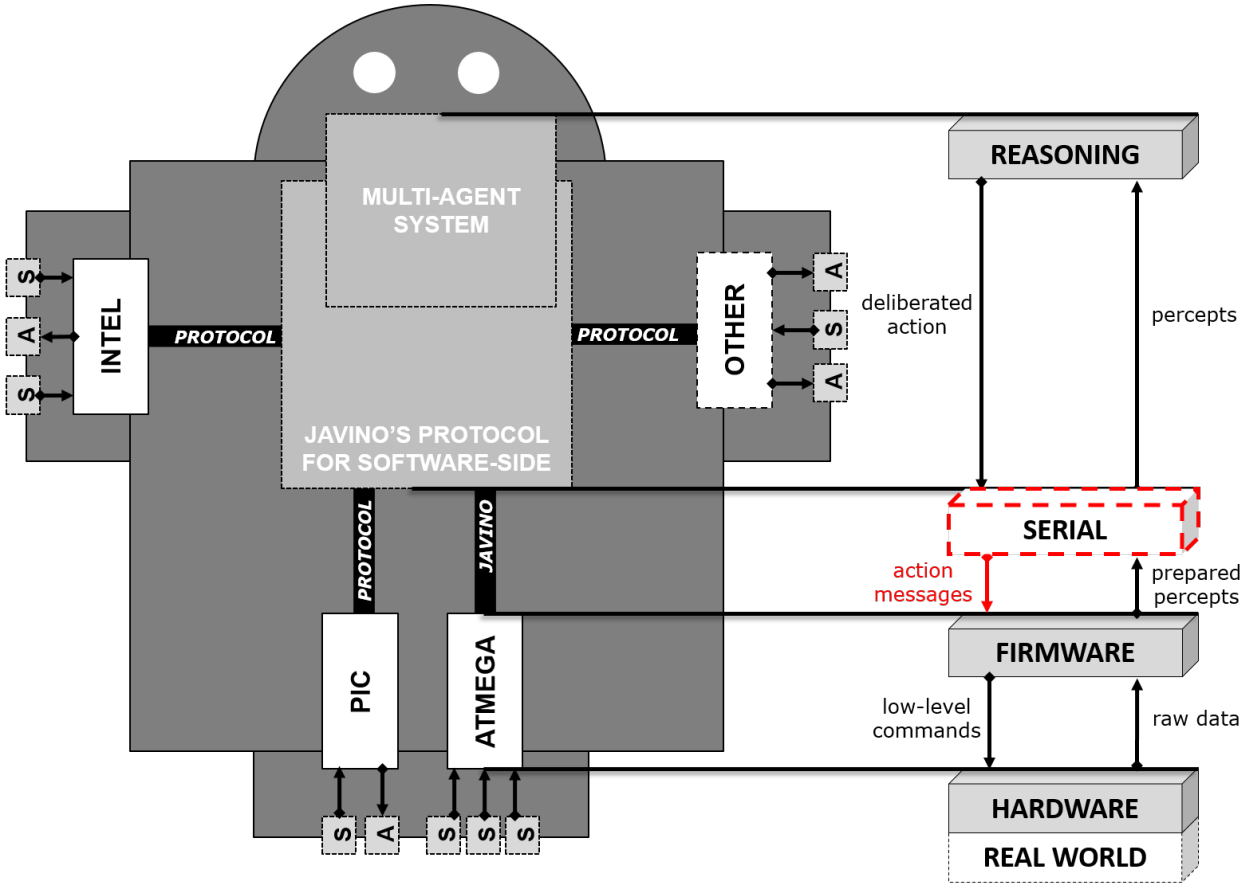
The agent can reason with perceptions that come directly from the real world.

Physical and Logical Architecture



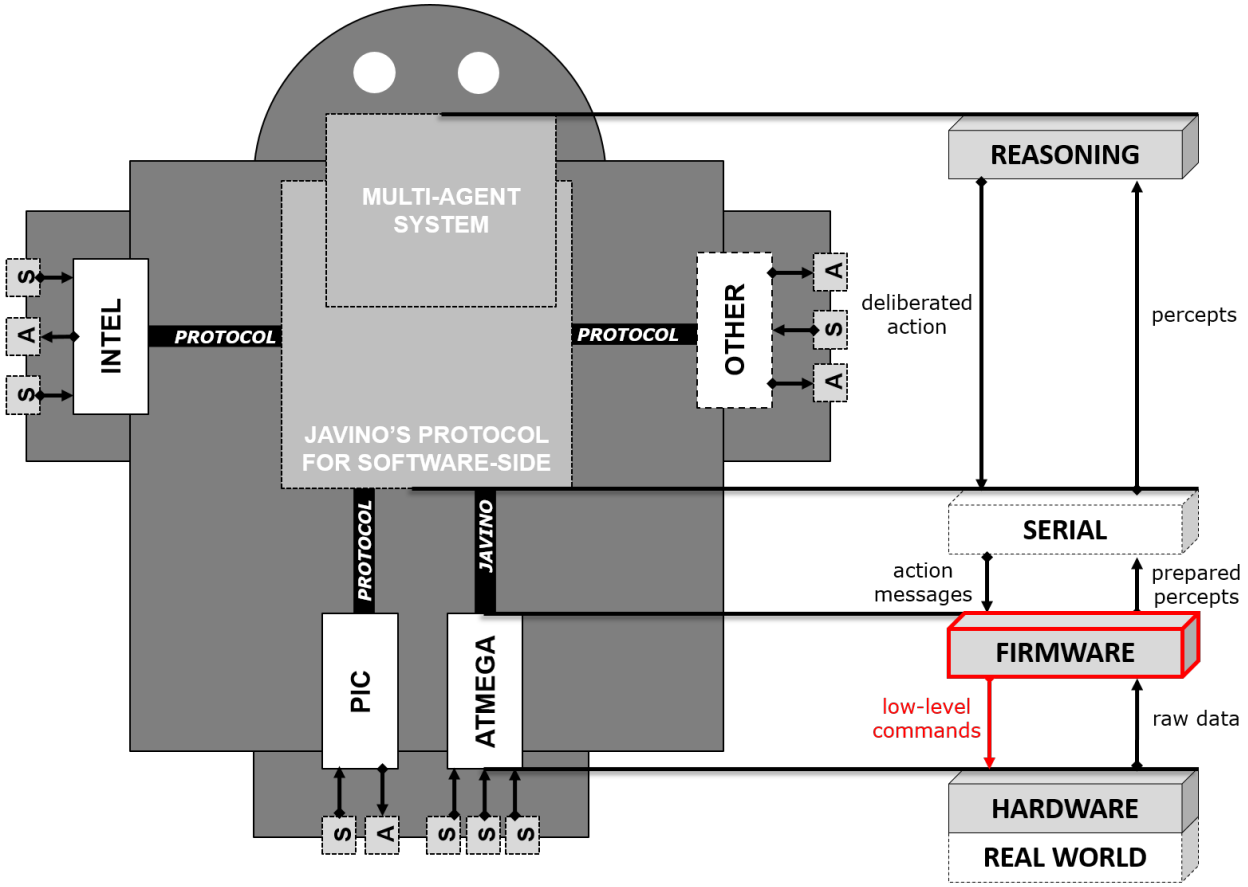
Then the agent deliberates if some action needs to be taken. In this case, a message is sent to the serial layer.

Physical and Logical Architecture



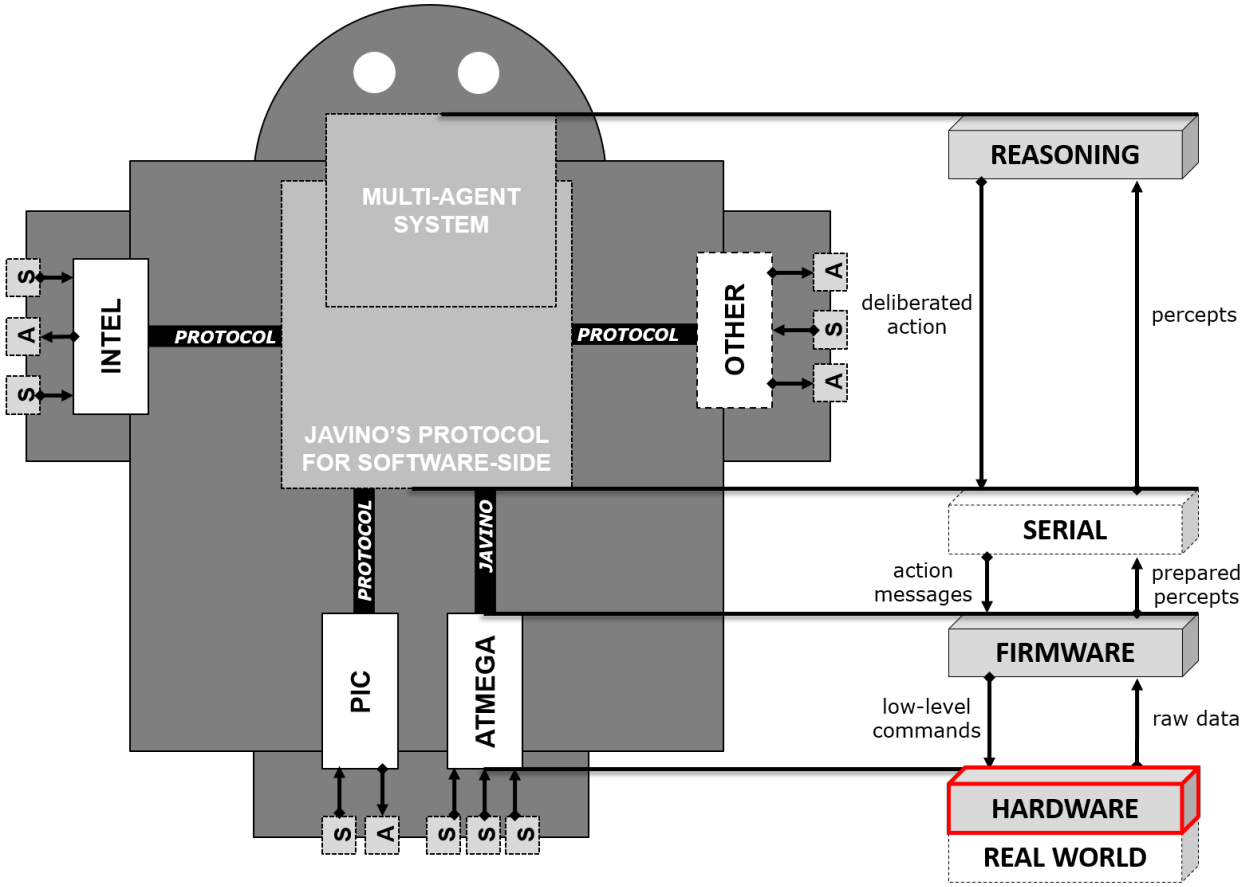
It redirects action messages to the microcontroller connected to the USB port identified in the message.

Physical and Logical Architecture



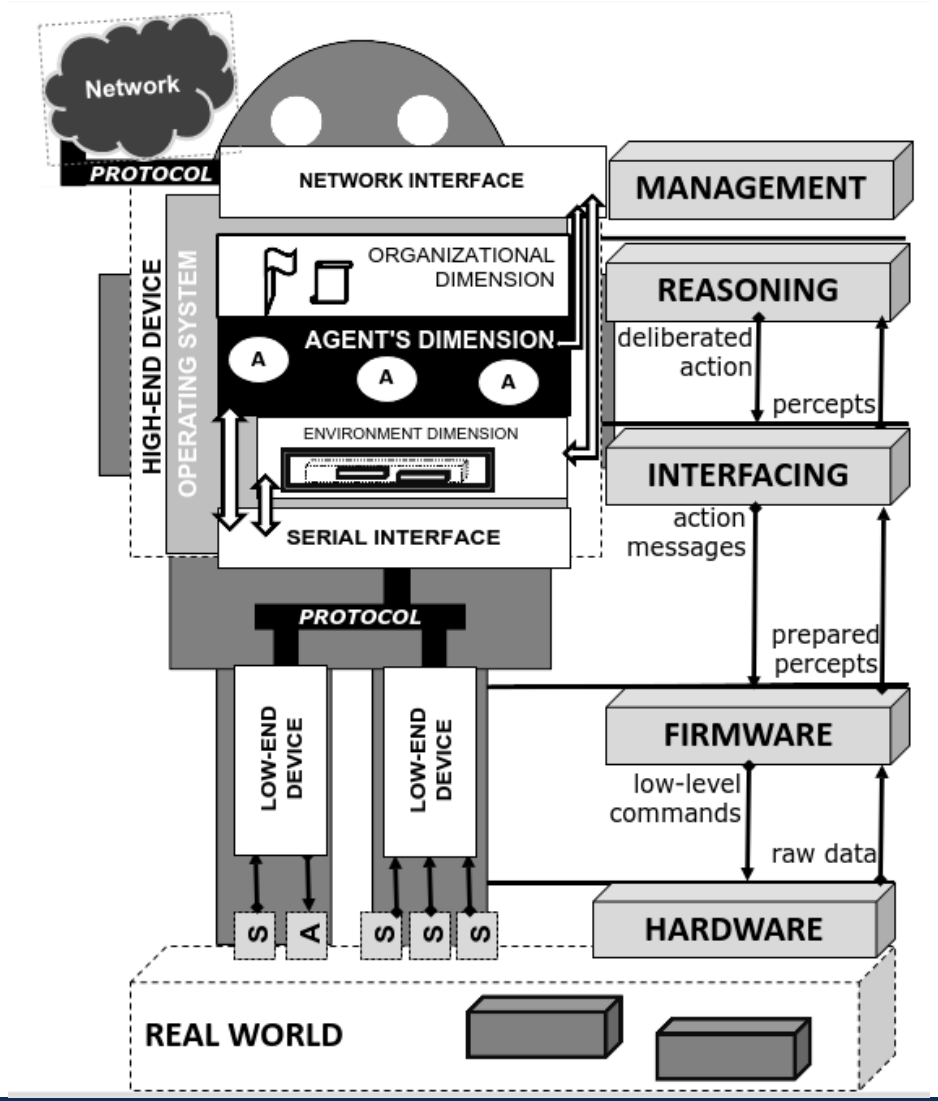
All possible actuator functions are programmed to be executed in response to messages from the serial port.

Physical and Logical Architecture



The actuator is activated.

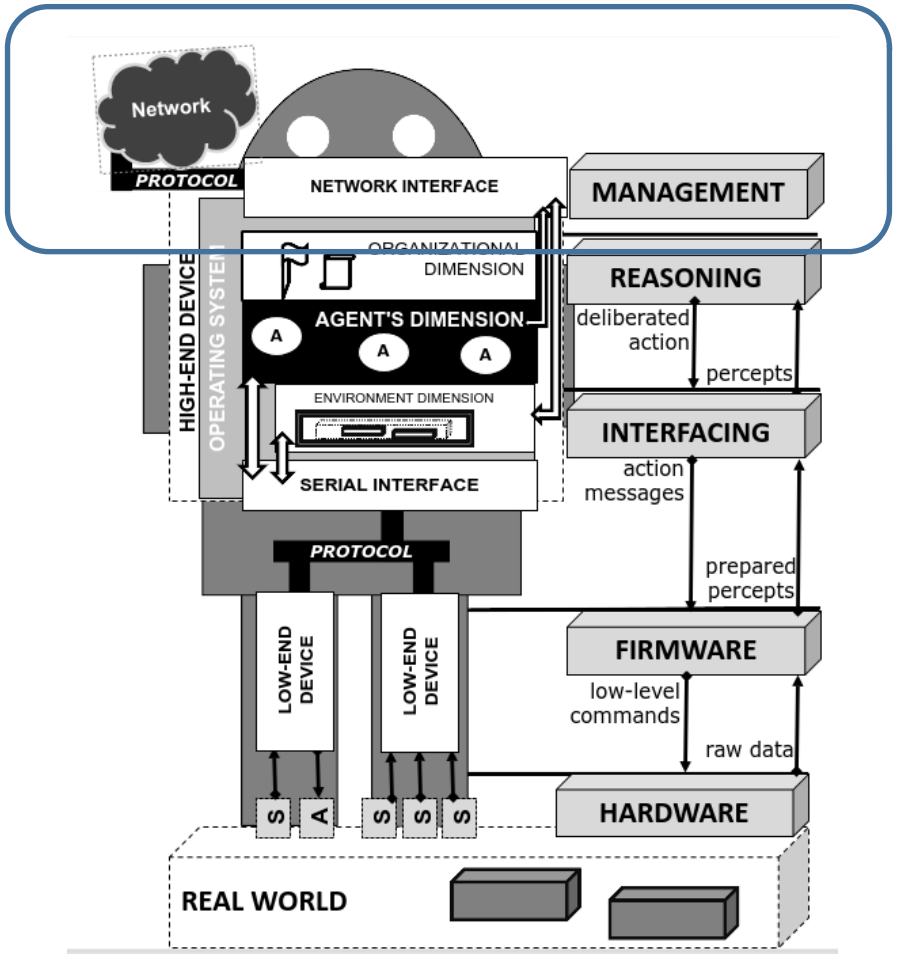
Physical and Logical Architecture



The system management.

Physical and Logical Architecture

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  - **reasoning**: where the MAS is hosted.
  - **management:** the operating system's resource management.



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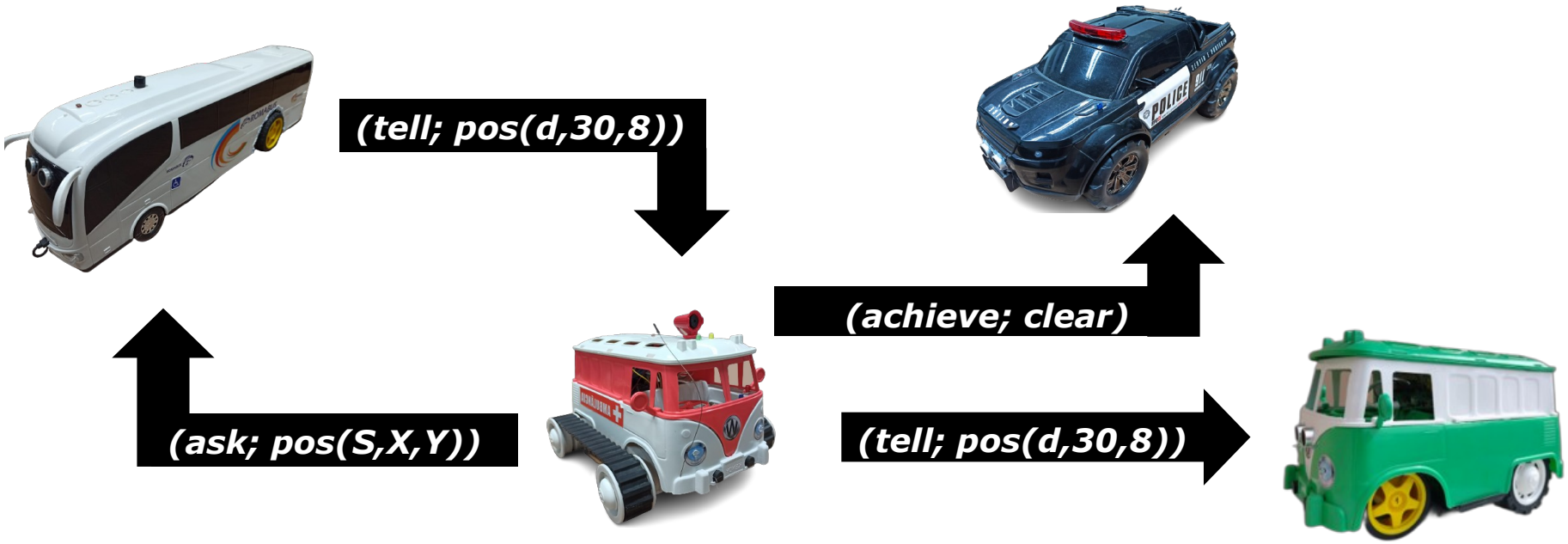






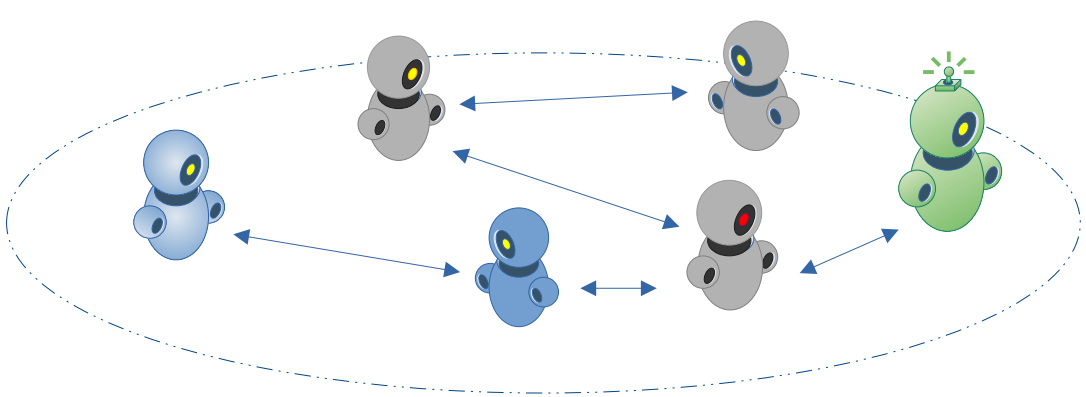
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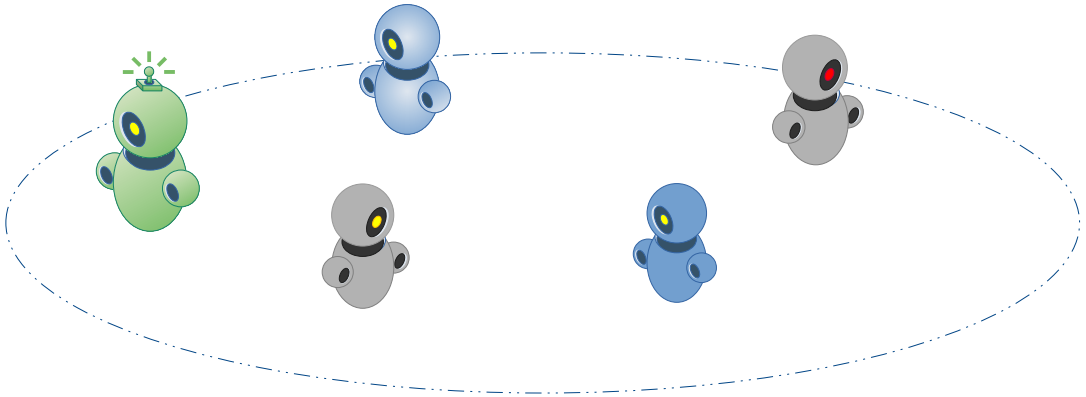
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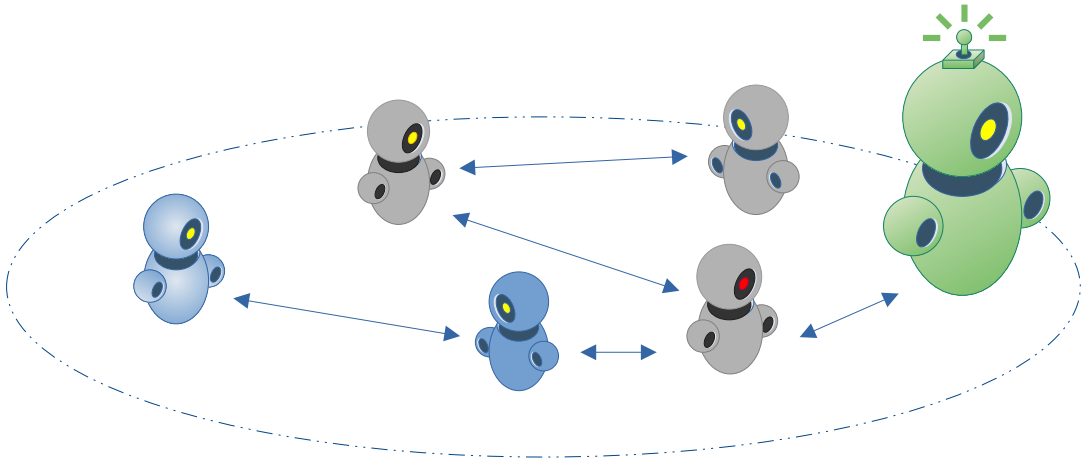
Princípio da Abertura com Comunicação

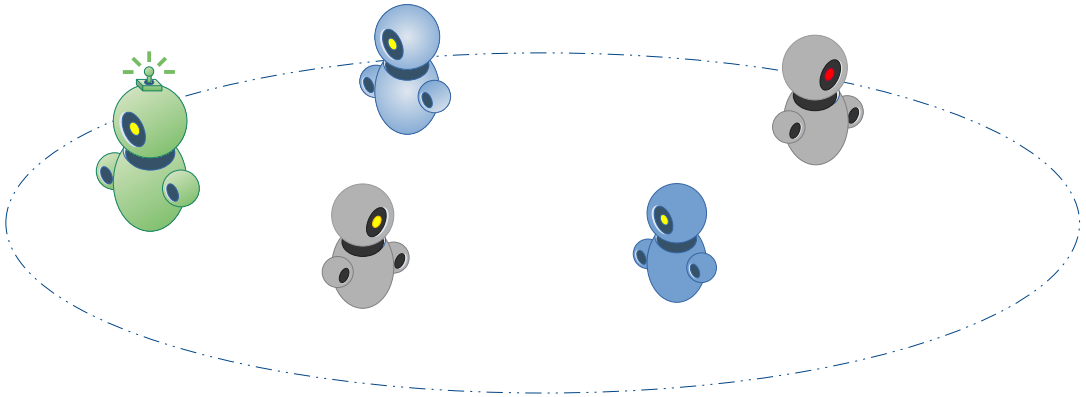




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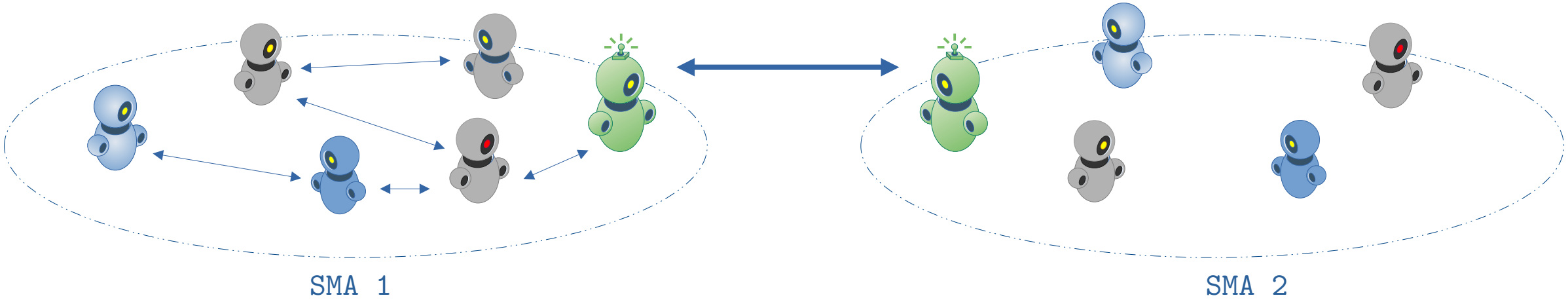
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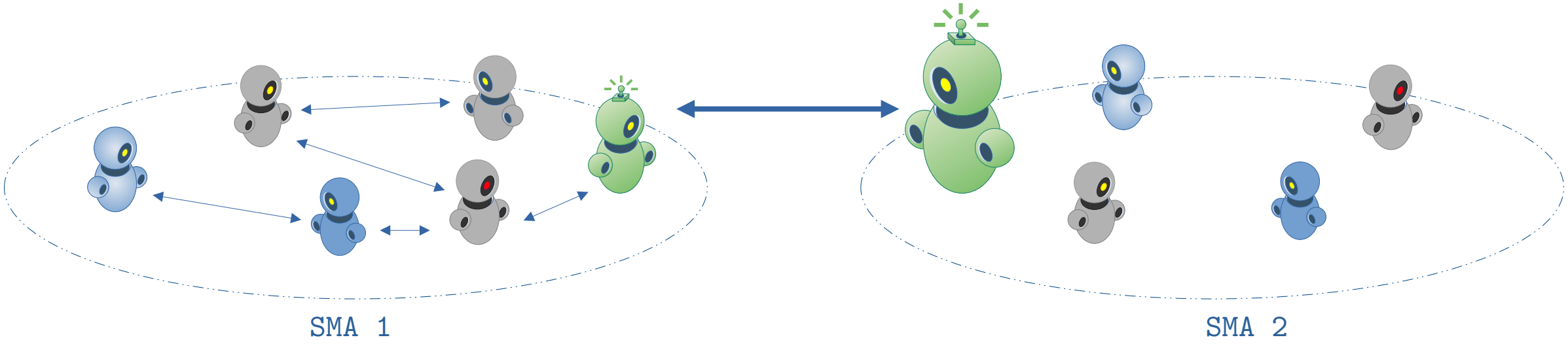
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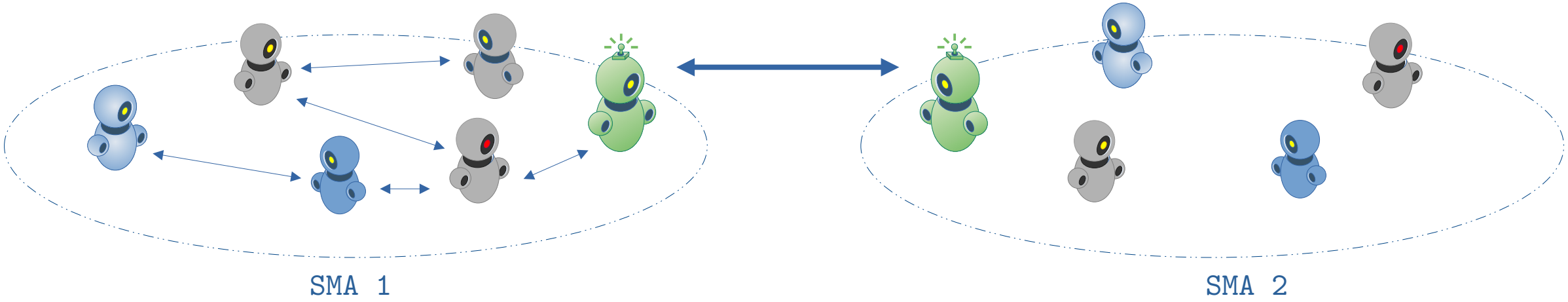
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Princípio da Abertura com Comunicação



PANTOJA, CARLOS; SOARES, HEDER DORNELES; VITERBO, JOSÉ; SEGHROUCHNI, AMAL EL FALLAH . An Architecture for the Development of Ambient Intelligence Systems Managed by Embedded Agents. In : The 30th International Conference on Software Engineering and Knowledge Engineering, 2018, San Francisco. 2018. p. 215.
Pantoja, Carlos Eduardo; VITERBO, JOSÉ; SEGHROUCHNI, AMAL EL-FALLAH. From Thing to Smart Thing : Towards an Architecture for Agent-Based AmI Systems. 13th KES International Conference, KES-AMSTA-2019. Smart Innovation, Systems and Technologies. 1ed.: Springer Singapore, 2020, v. 148, p. 57-67.

Princípio da Abertura com Comunicação



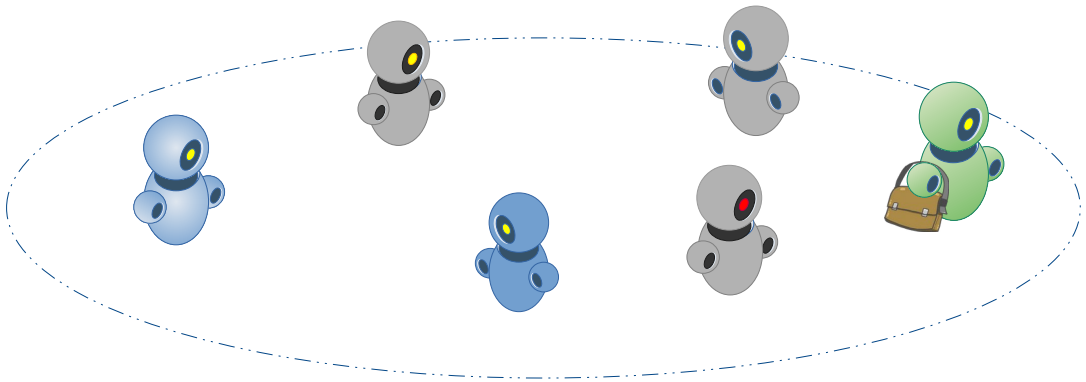
PANTOJA, CARLOS; SOARES, HEDER DORNELES; VITERBO, JOSÉ; SEGHROUCHNI, AMAL EL FALLAH . An Architecture for the Development of Ambient Intelligence Systems Managed by Embedded Agents. In : The 30th International Conference on Software Engineering and Knowledge Engineering, 2018, San Francisco. 2018. p. 215.
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Agentes Móveis

Os agentes móveis [Azzouzi et al., 2010] são capazes de movimentarem-se entre diferentes SMA.

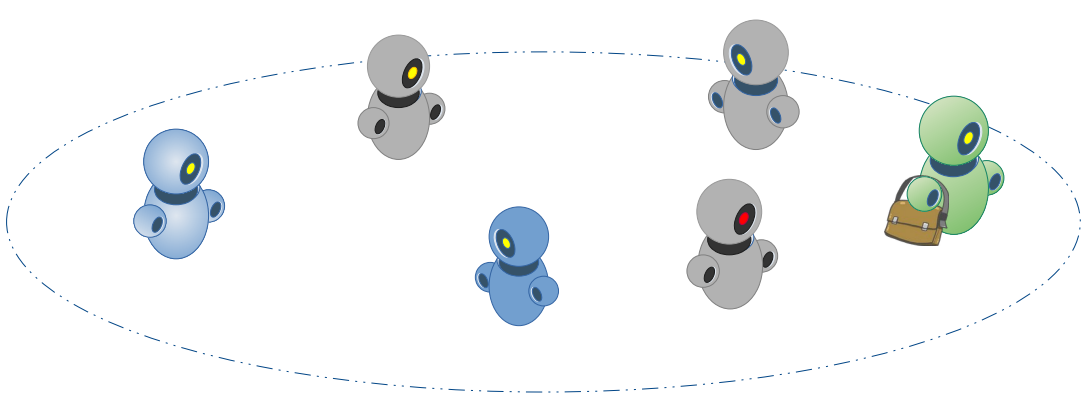
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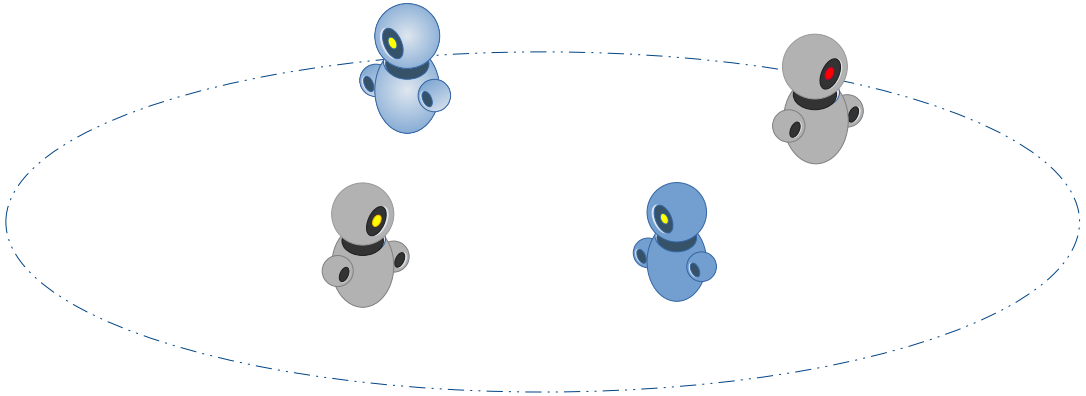


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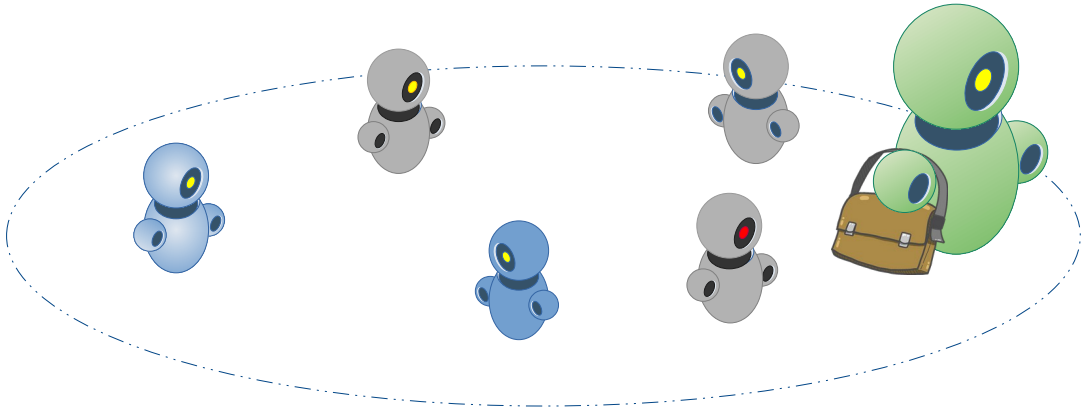
SMA 1



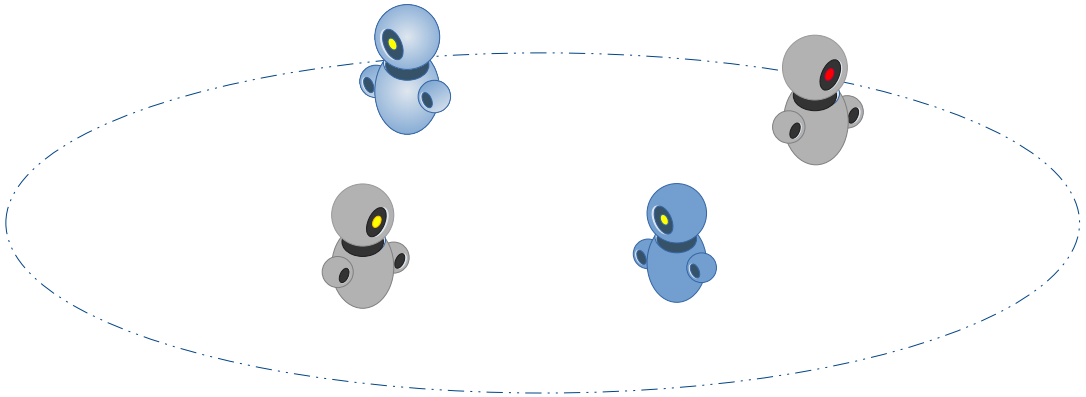
SMA 2

Agentes Móveis

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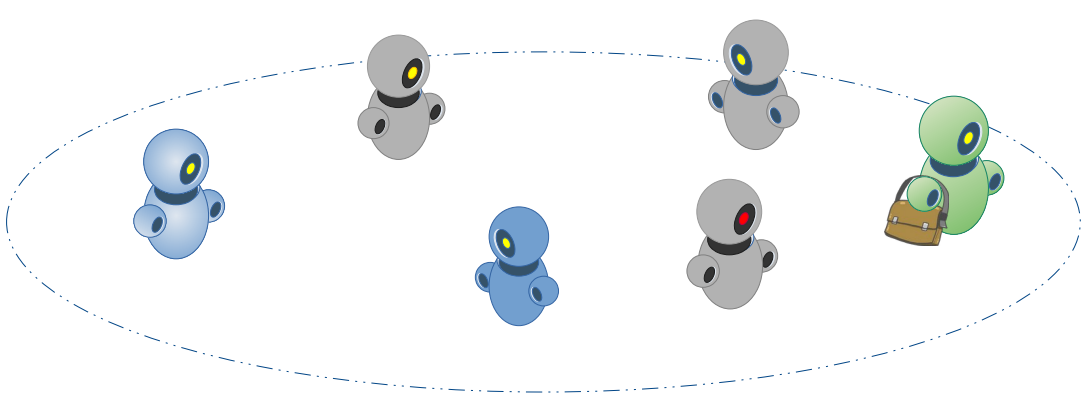
SMA 1



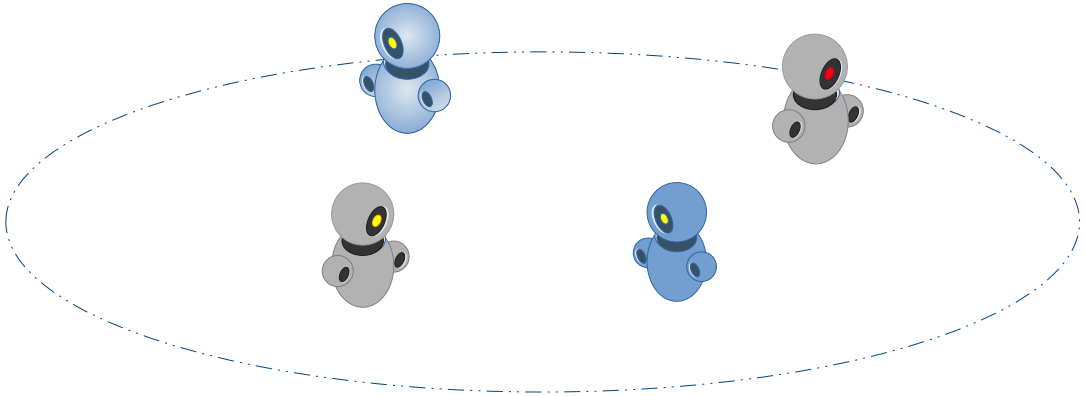
SMA 2

Agentes Móveis

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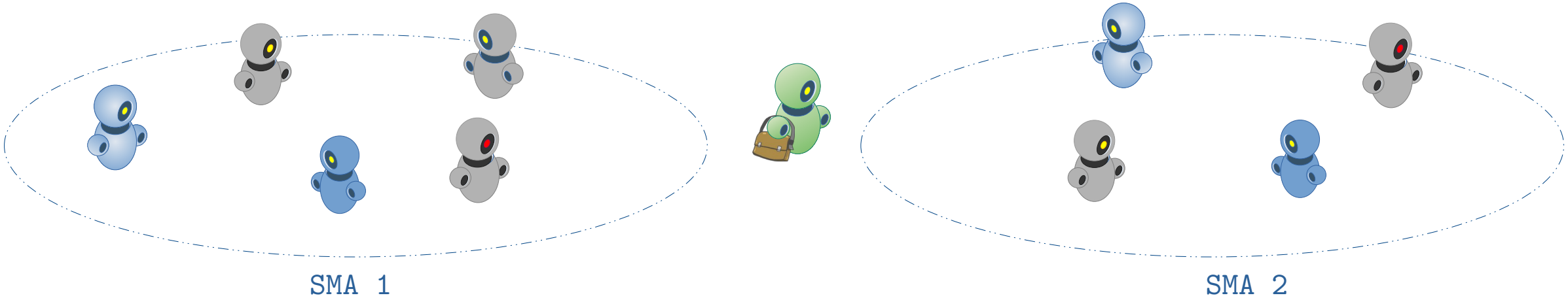
SMA 1



SMA 2

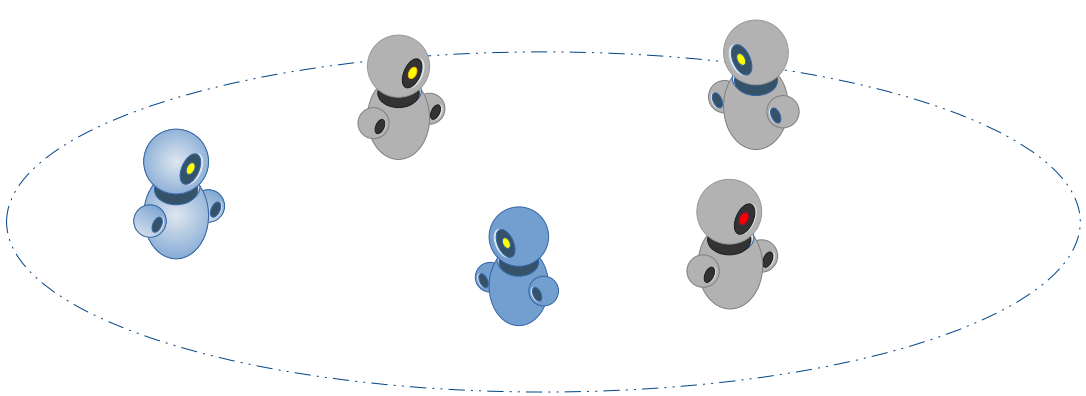
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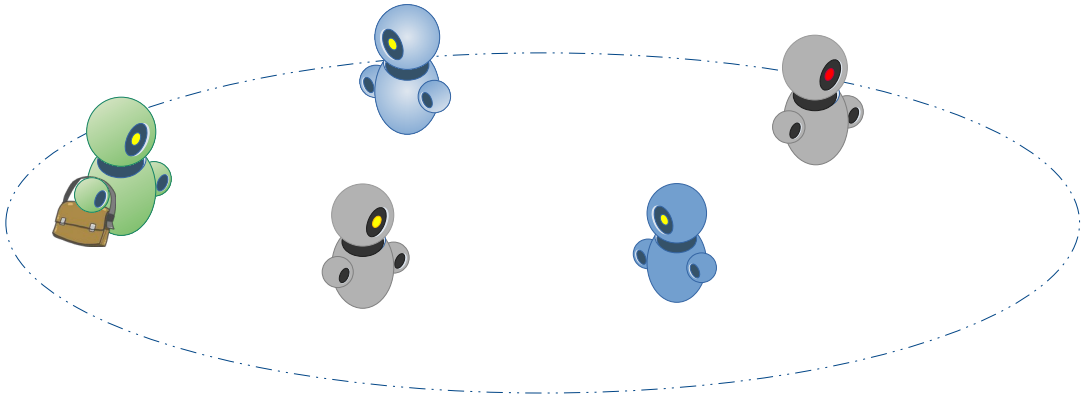


Agentes Móveis

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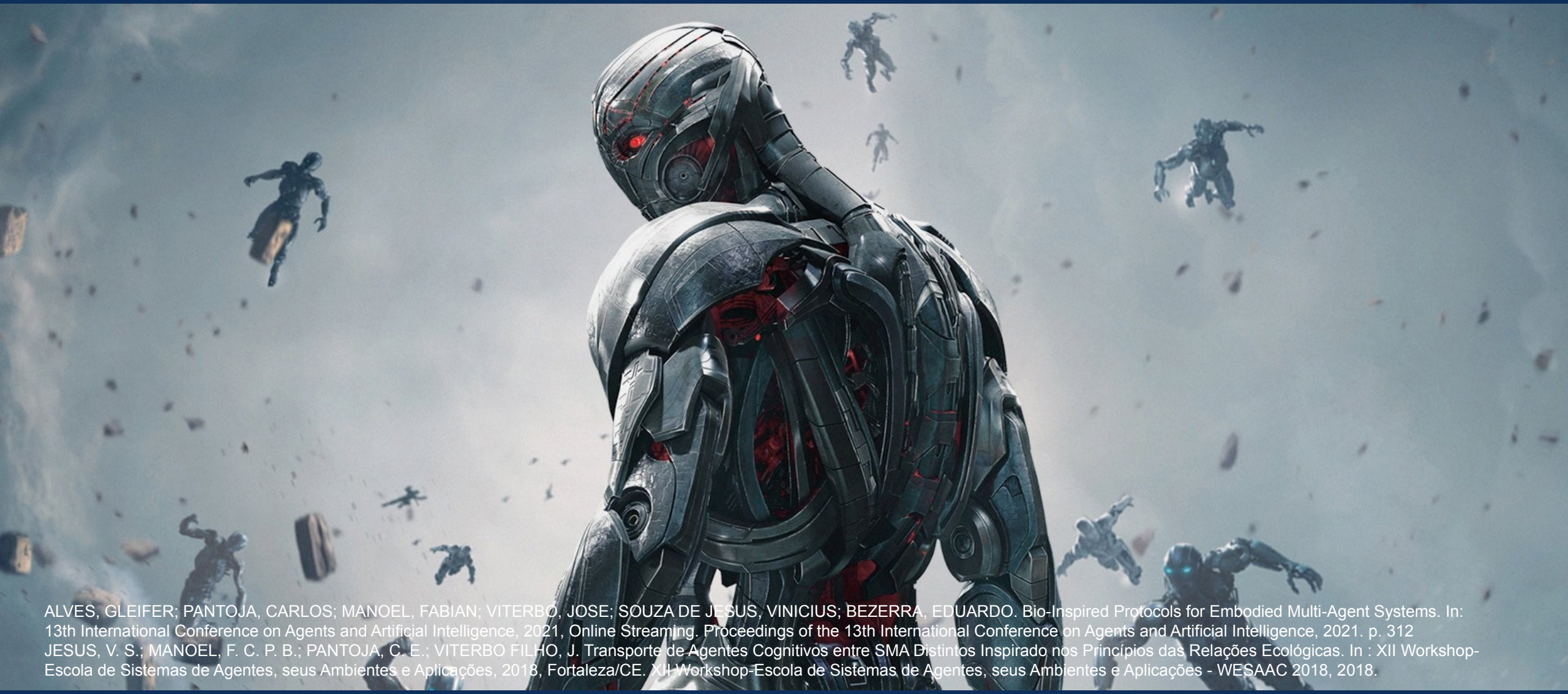


SMA 1



SMA 2

The Road So Far



ALVES, GLEIFER; PANTOJA, CARLOS; MANOEL, FABIAN; VITERBO, JOSE; SOUZA DE JESUS, VINICIUS; BEZERRA, EDUARDO. Bio-Inspired Protocols for Embodied Multi-Agent Systems. In: 13th International Conference on Agents and Artificial Intelligence, 2021, Online Streaming. Proceedings of the 13th International Conference on Agents and Artificial Intelligence, 2021. p. 312
JESUS, V. S.; MANOEL, F. C. P. B.; PANTOJA, C. E.; VITERBO FILHO, J. Transporte de Agentes Cognitivos entre SMA Distintos Inspirado nos Princípios das Relações Ecológicas. In : XII Workshop-Escola de Sistemas de Agentes, seus Ambientes e Aplicações, 2018, Fortaleza/CE. XII Workshop-Escola de Sistemas de Agentes, seus Ambientes e Aplicações - WESAAC 2018, 2018.

Bio-Inspired

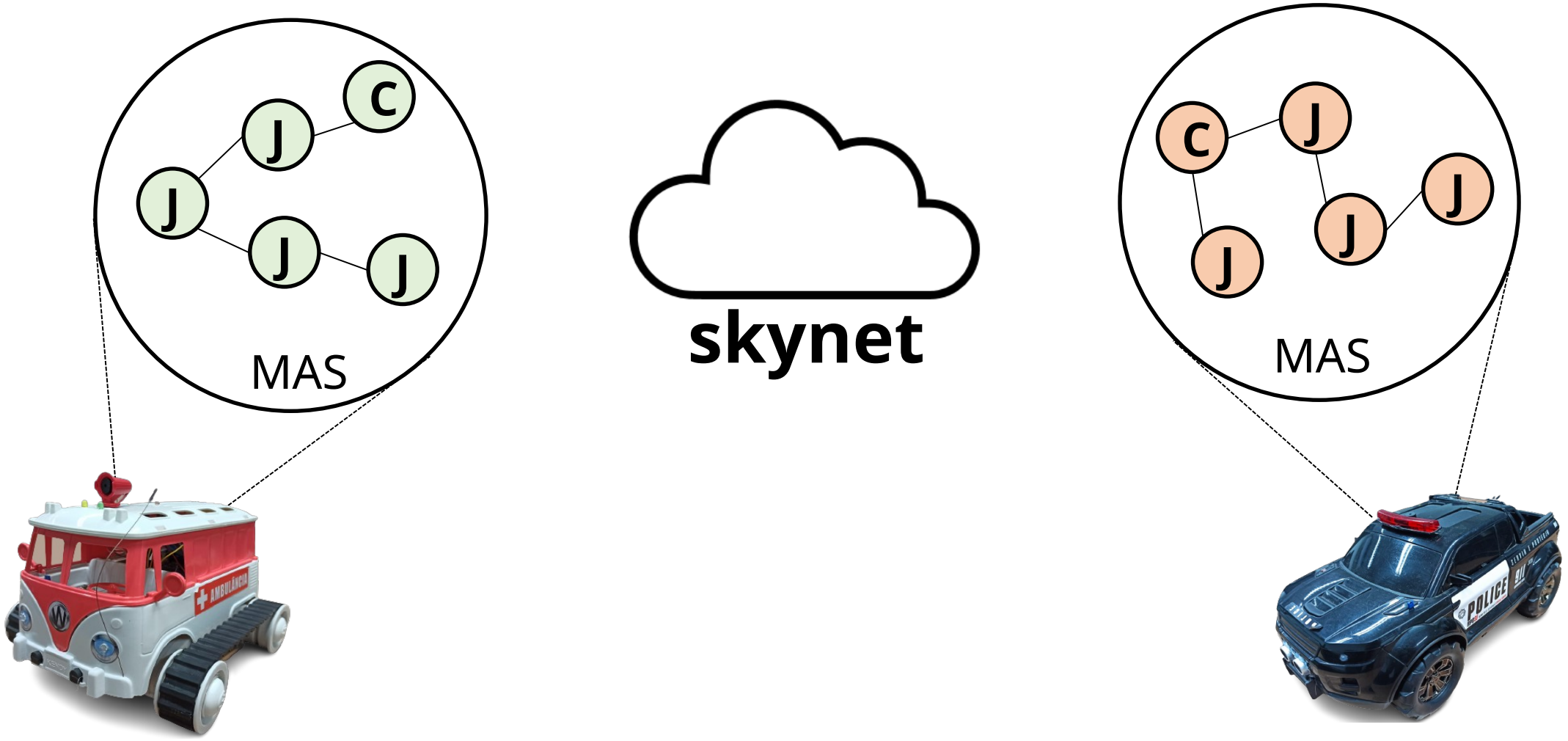
O protocolo de transferência de agentes prevê três possíveis relações entre o agente móvel com o novo SMA.

Mutualismo

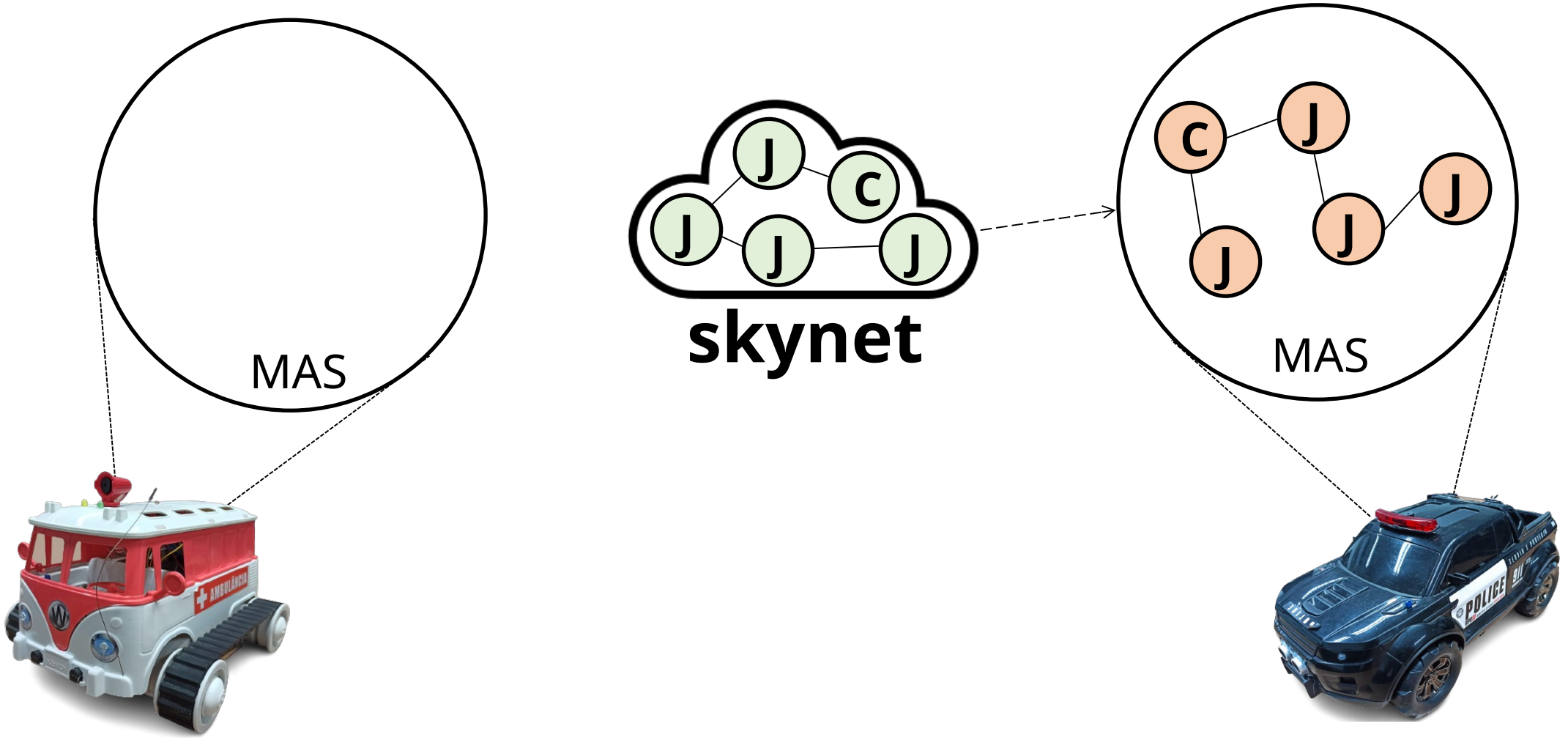
Inquilinismo

Predatismo

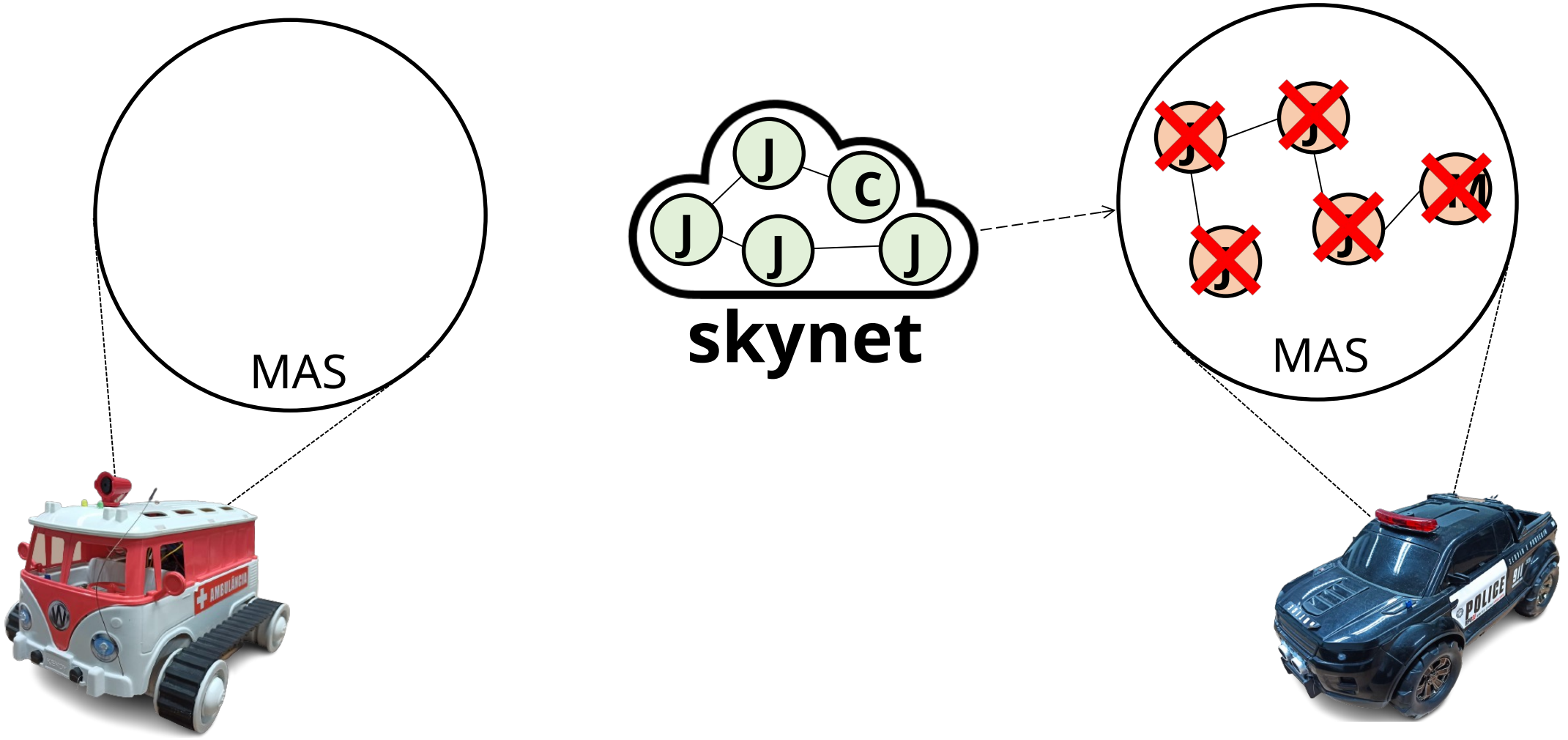
Bio-Inspired: Predation



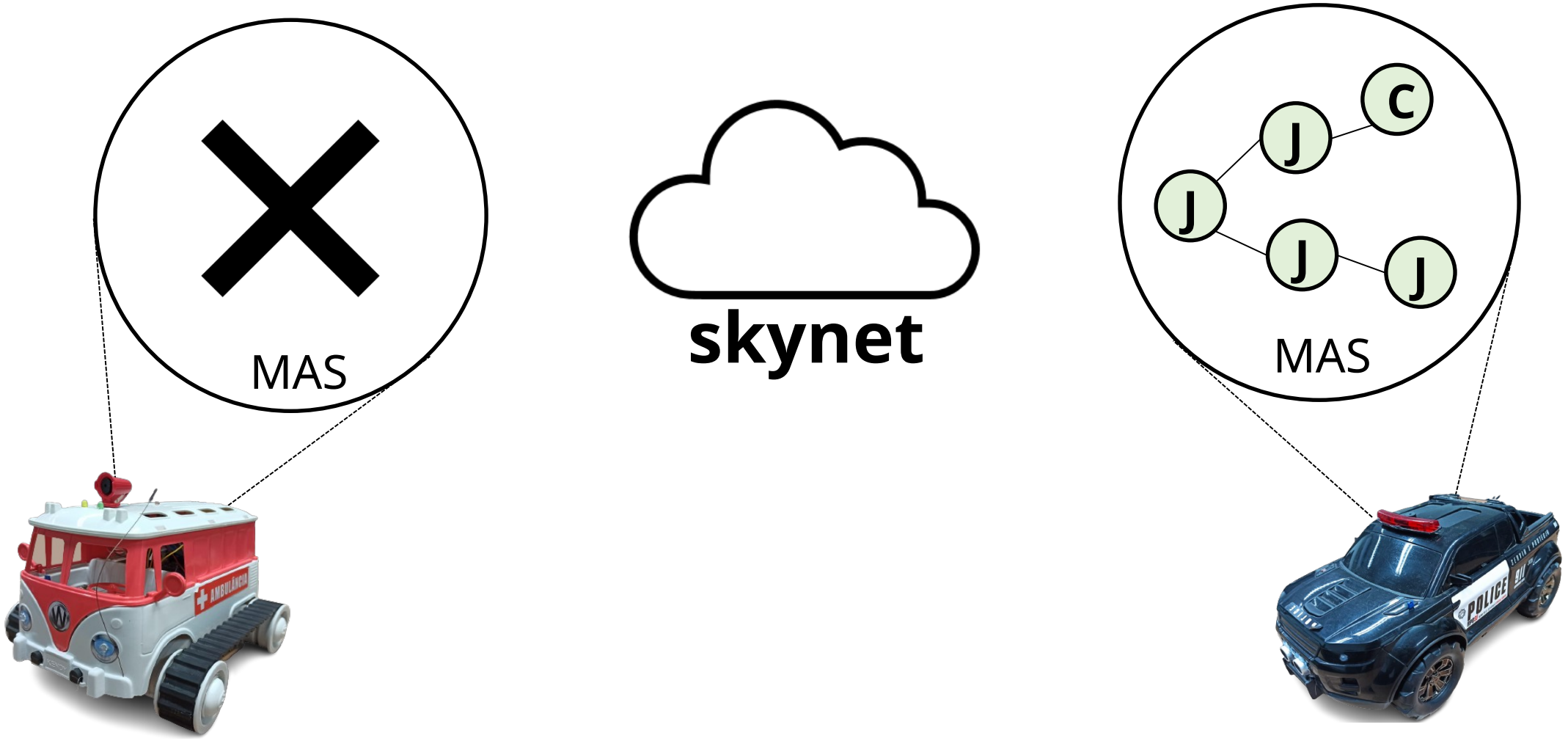
Bio-Inspired: Predation



Bio-Inspired: Predation



Bio-Inspired: Predation



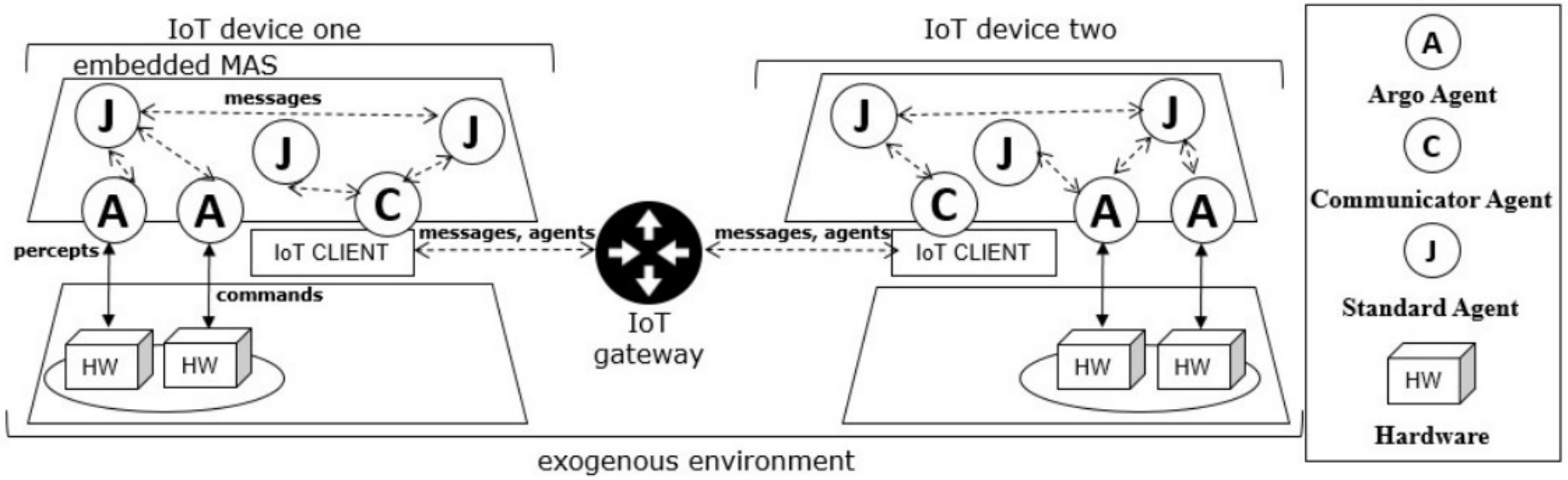
Princípio de Abertura com Comunicação

	Totally Closed MAS	Limited Open MAS	Closed MAS	Open MAS
Entrance	✗	✓	✗	✓
Communication	✗	✗	✓	✓

Jesus, V.S.d., Lazarin, N.M., Pantoja, C.E. et al. A middleware for providing communicability to Embedded MAS based on the lack of connectivity. Artif Intell Rev 56 (Suppl 3), 2971–3001 (2023). https://doi.org/10.1007/s10462-023-10596-z

Jason Embedded

It is a spin-off Version of Jason for IoT and Embedded Multiagent Systems.



PANTOJA, Carlos Eduardo; JESUS, Vinicius Souza de; LAZARIN, Nilson Mori; VITERBO, José. A Spin-off Version of Jason for IoT and Embedded Multi-Agent Systems. In: BRAZILIAN CONFERENCE ON INTELLIGENT SYSTEMS (BRACIS), 12. , 2023, Belo Horizonte/MG. Anais [...]. Porto Alegre: Sociedade Brasileira de Computação, 2023 . p. 382-396. ISSN 2643-6264.

Agentes Comunicadores: Ações Internas

.sendOut(agentUuid, force, message)

envia uma mensagem de um agente comunicador para outro comunicador de um SMA distinto.

.connectCN(servidor, porta, UUID)

se conecta ao servidor IoT com um id específico para o agente.

.disconnectCN

desconecta do servidor IoT atualmente conectado.

Communicator: Ações Internas

.moveOut(agentUuid, predation|inquilinism)

move os agentes para um SMA.

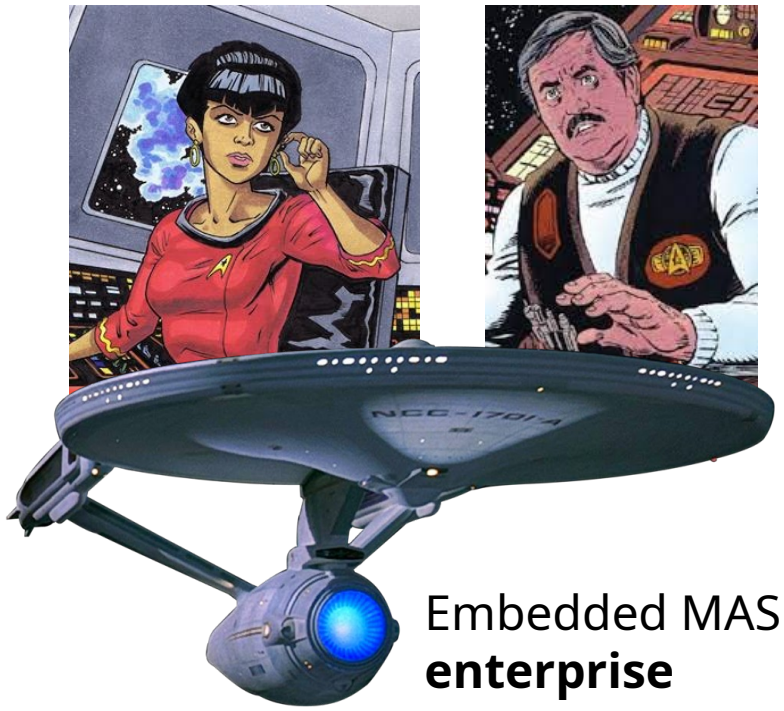
.moveOut(agentUuid, mutualism, agent)

envia um agente específico para outro SMA.

Exemplo: Bio-Inspired Protocol

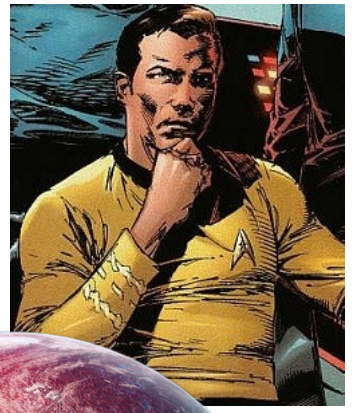
agent **uhura**

agent **scott**



Embedded MAS
enterprise

agent **kirk**



Embedded MAS
andoria



agent **spok**

Exemplo: Bio-Inspired Protocol

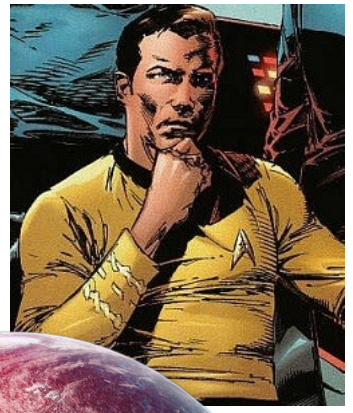




Exemplo: Bio-Inspired Protocol



Computer, Commander Montgomery Scott, Chief Engineering Office!





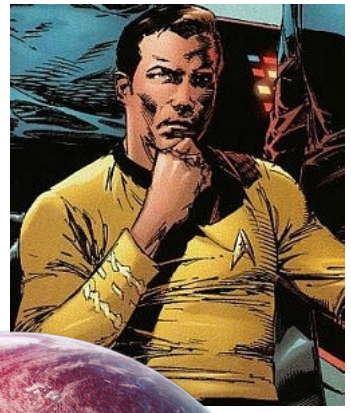


Exemplo: Bio-Inspired Protocol



Computer, Commander Montgomery Scott, Chief Engineering Office!

Kirk to Scotty… Beam us up!





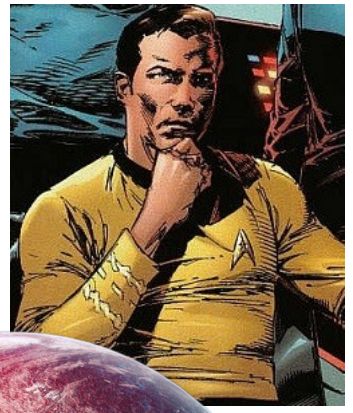
Exemplo: Bio-Inspired Protocol



Computer, Commander Montgomery Scott, Chief Engineering Office!

Transporter ready for Kirk and Spok.

Kirk to Scotty... Beam us up!







Exemplo: Bio-Inspired Protocol



Computer, Commander James T. Kirk, Enterprise's Captain! I'm aboard!



Exemplo: Bio-Inspired Protocol



```
.connectCN("skynet.chon.group",3273,"07ba9e4a-d539-4a0e-8c14-4ac336476858").
```

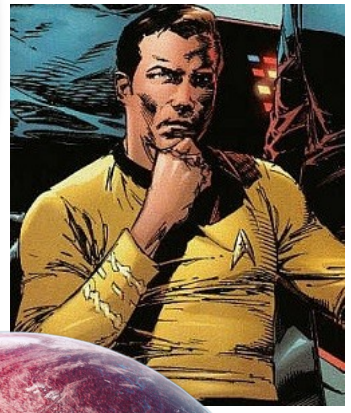


Exemplo: Bio-Inspired Protocol



.connectCN("skynet.chon.group",3273,"07ba9e4a-d539-4a0e-8c14-4ac336476858").

.sendOut(scott, tell, beam_us_up)



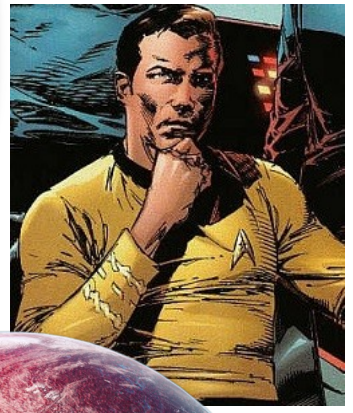


Exemplo: Bio-Inspired Protocol



.connectCN("skynet.chon.group",3273,"07ba9e4a-d539-4a0e-8c14-4ac336476858").

.sendOut(scott, tell, beam_us_up)



.sendOut(kirk,tell, energizing);



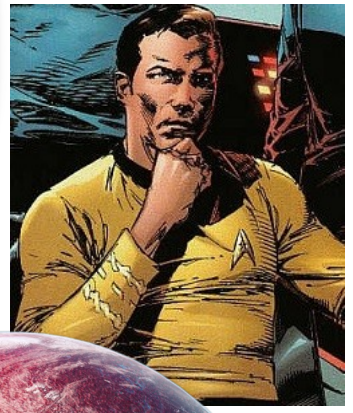


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.sendOut(scott, tell, beam_us_up)



.moveOut(scott, inquilinism).

.sendOut(kirk,tell, energizing);





Exemplo: Bio-Inspired Protocol



Computer, Commander James T. Kirk, Enterprise's Captain! I'm aboard!



Communicator Concierge: Ações Internas

.guidance(input|output,
communication|migration|all,
agent's address,
all|kqml|bioinsp,
accept|drop)

defines if a communication or migration can be accepted or rejected.

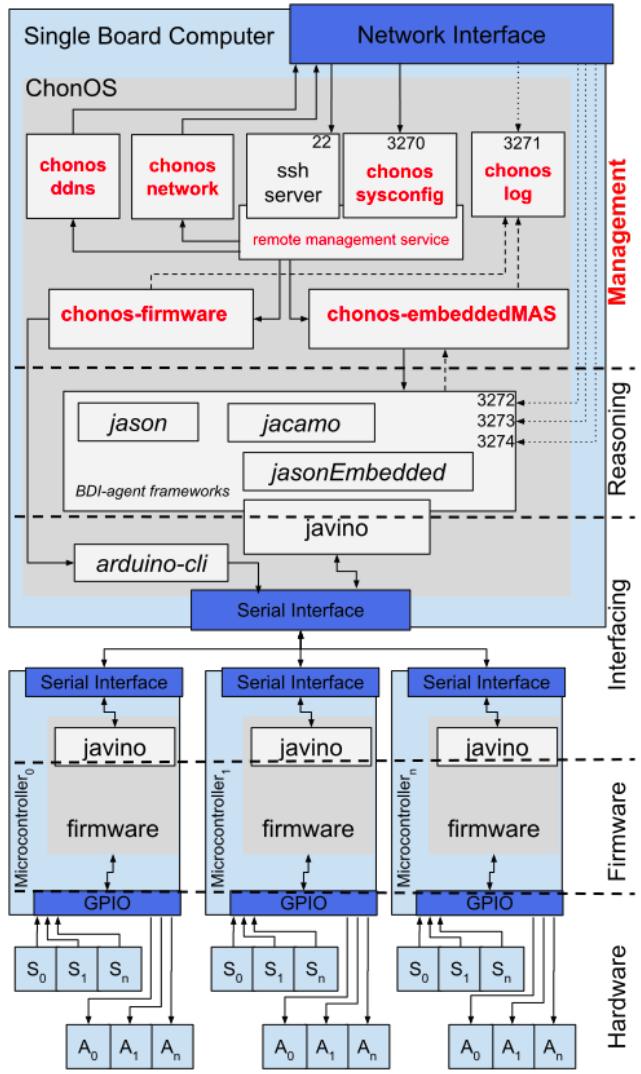
Communicator Concierge: Ações Internas

```
.policy(input|output,
        communication|migration|all,
        all|kqml|bioinsp,
        accept|drop)
```

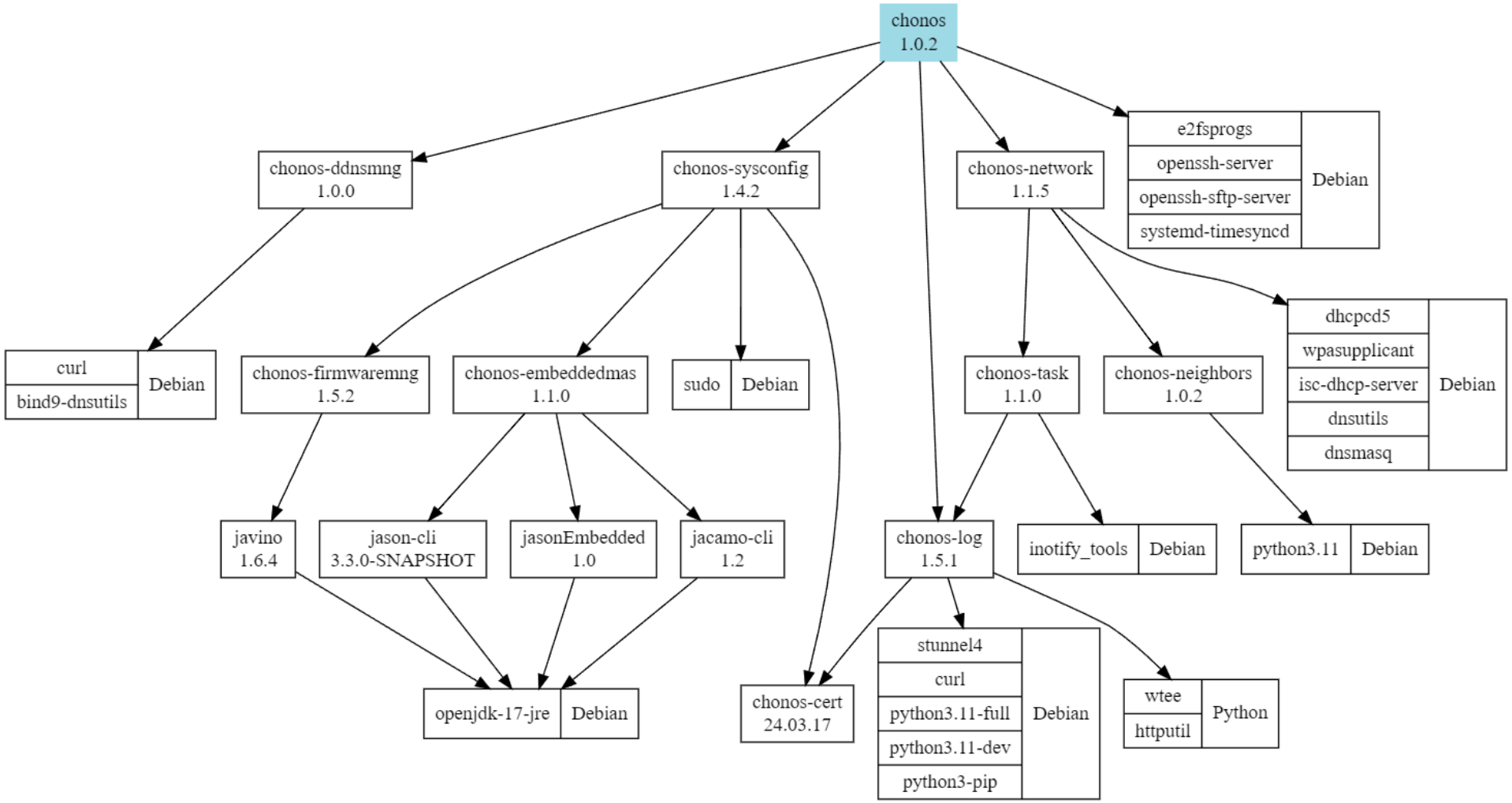
Defines the accepting or rejecting communication or migration if there isn’t applicable guidance.

chonOS

ChonOS is a ready-to-use specific-purpose GNU/Linux distribution to facilitate the development process and use of BDI-based Embedded MAS.



chonOS


chonos
1.0.2
chonos-ddnsmng
1.0.0
chonos-sysconfig
1.4.2
chonos-network
1.1.5
e2fsprogs
openssh-server
openssh-sftp-server
systemd-timesyncd
Debian
curl
bind9-dnsutils
Debian
chonos-firmwaremng
1.5.2
chonos-embeddedmas
1.1.0
sudo
Debian
chonos-task
1.1.0
chonos-neighbors
1.0.2
dhcpcd5
wpasupplicant
isc-dhcp-server
dnsutils
dnsmasq
Debian
javino
1.6.4
jason-cli
3.3.0-SNAPSHOT
jasonEmbedded
1.0
jacamo-cli
1.2
chonos-log
1.5.1
inotify_tools
Debian
python3.11
Debian
openjdk-17-jre
Debian
chonos-cert
24.03.17
stunnel4
curl
python3.11-full
python3.11-dev
python3-pip
Debian
wtee
httputil
Python

The Road So Far

	Technology	Reference
Management	chonOS	[61]
Reasoning	Jason Embedded	[16, 66, 60]
Interfacing	Javino	[76]
	Javic	[12]
	Virtual Ports	[63]
Firmware	Javino	[76]
Hardware	chonBot	[61]

[12] Guinelli, Joao Victor, Pantoja, Carlos, A Middleware for Using PIC Microcontrollers and Jason Framework for Programming Multi-Agent Systems, In: Anais do WPCCG. Ponta Grossa, 2016;1. http://www.wpccg.pro.br/volume001.html

[16] PANTOJA, Carlos Eduardo; JESUS, Vinicius Souza de; LAZARIN, Nilson Mori; VITERBO, José. A Spin-off Version of Jason for IoT and Embedded Multi-Agent Systems. In: BRAZILIAN CONFERENCE ON INTELLIGENT SYSTEMS (BRACIS), 12. , 2023, Belo Horizonte/MG. Anais [...]. Porto Alegre: Sociedade Brasileira de Computação, 2023 . p. 382-396. ISSN 2643-6264.

[60] Lazarin, Nilson Mori, de Souza, João Pedro Bernardo, Pantoja, Carlos Eduardo, Alexandre, Tielle, Viterbo, Jose, Human and BDI-Agent Interaction via KQML Messages over IMAP and SMTP, In: Advances in Practical Applications of Agents, Multi-Agent Systems, and Digital Twins: The PAAMS Collection. Cham, 2025;:172–183. DOI:10.1007/978-3-031-70415-4 15.

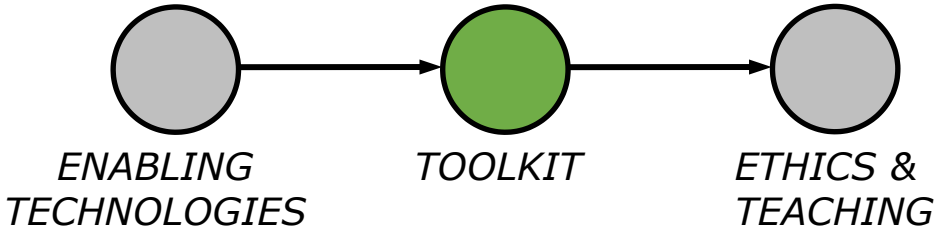
[61] Lazarin, Nilson Mori, Pantoja, Carlos Eduardo, Viterbo, José, Towards a Toolkit for Teaching AI Supported by Robotic-agents: Proposal and First Impressions, In: Anais do XXXI Workshop sobre Educação em Computação (WEI 2023). João Pessoa, 2023;:20–29. DOI:10.5753/wei.2023.229753.

[63] Freitas, Bruno, Lazarin, Nilson, Pantoja, Carlos, A Proposal for a Serial Port Emulator for Embedded Multi-Agent Systems, In: Proceedings of the 17th Workshop-School on Agents, Environments, and Applications. Porto Alegre, RS, Brasil, 2023;:55–66. DOI:10.5753/wesaac.2023.33437

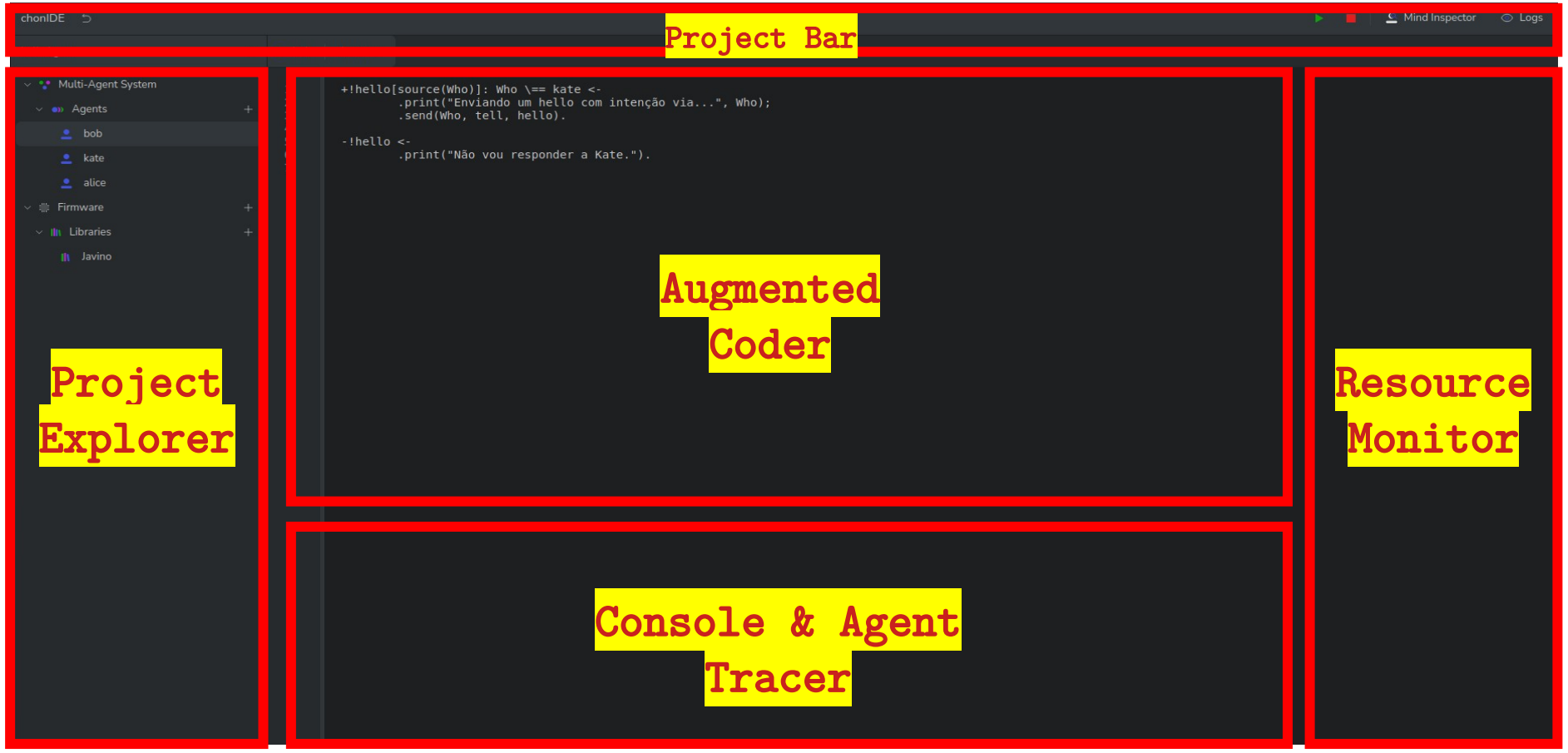
[66] Mori Lazarin, Nilson, Machado Coelho, Igor, Pantoja, Carlos Eduardo, Viterbo, José. Velluscinum: A Middleware for Using Digital Assets in Multi-agent Systems, In: Advances in Practical Applications of Agents, Multi-Agent Systems, and Cognitive Mimetics. The PAAMS Collection. Cham, 2023;13955:200–212. DOI:10.1007/978-3-031-37616-0 17.

[76] LAZARIN, N. M.; PANTOJA, C. E. A Robotic-agent Platform For Embedding Software Agents using Raspberry Pi and Arduino Boards. In : Workshop-Escola de Sistemas de Agentes, Seus Ambientes e Aplicações, 2015, Rio de Janeiro. Anais do IX Workshop-Escola de Sistemas de Agentes, seus Ambientes e Aplicações - WESAAC 2015, 2015.

The Road So Far

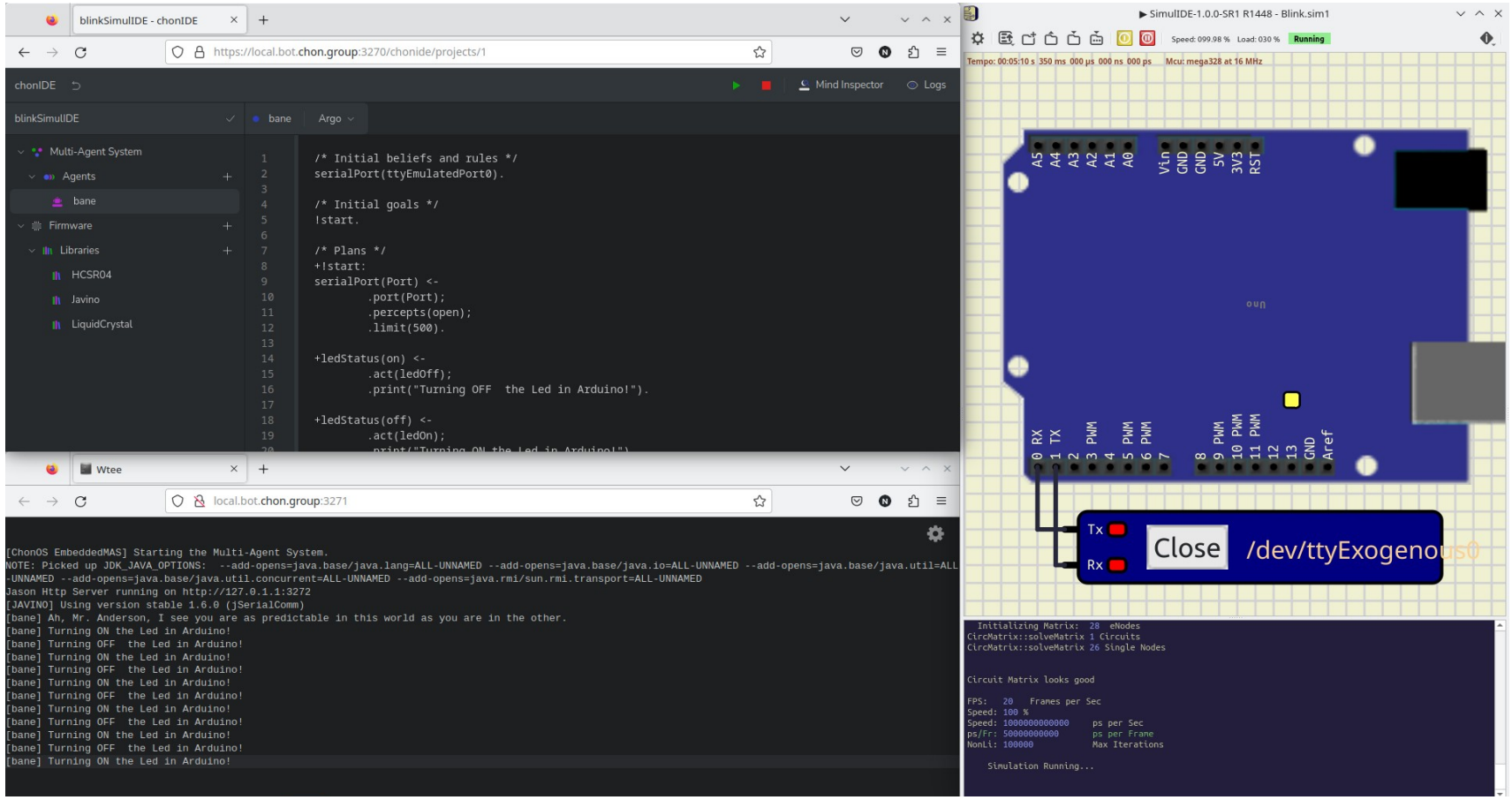


The Road So Far

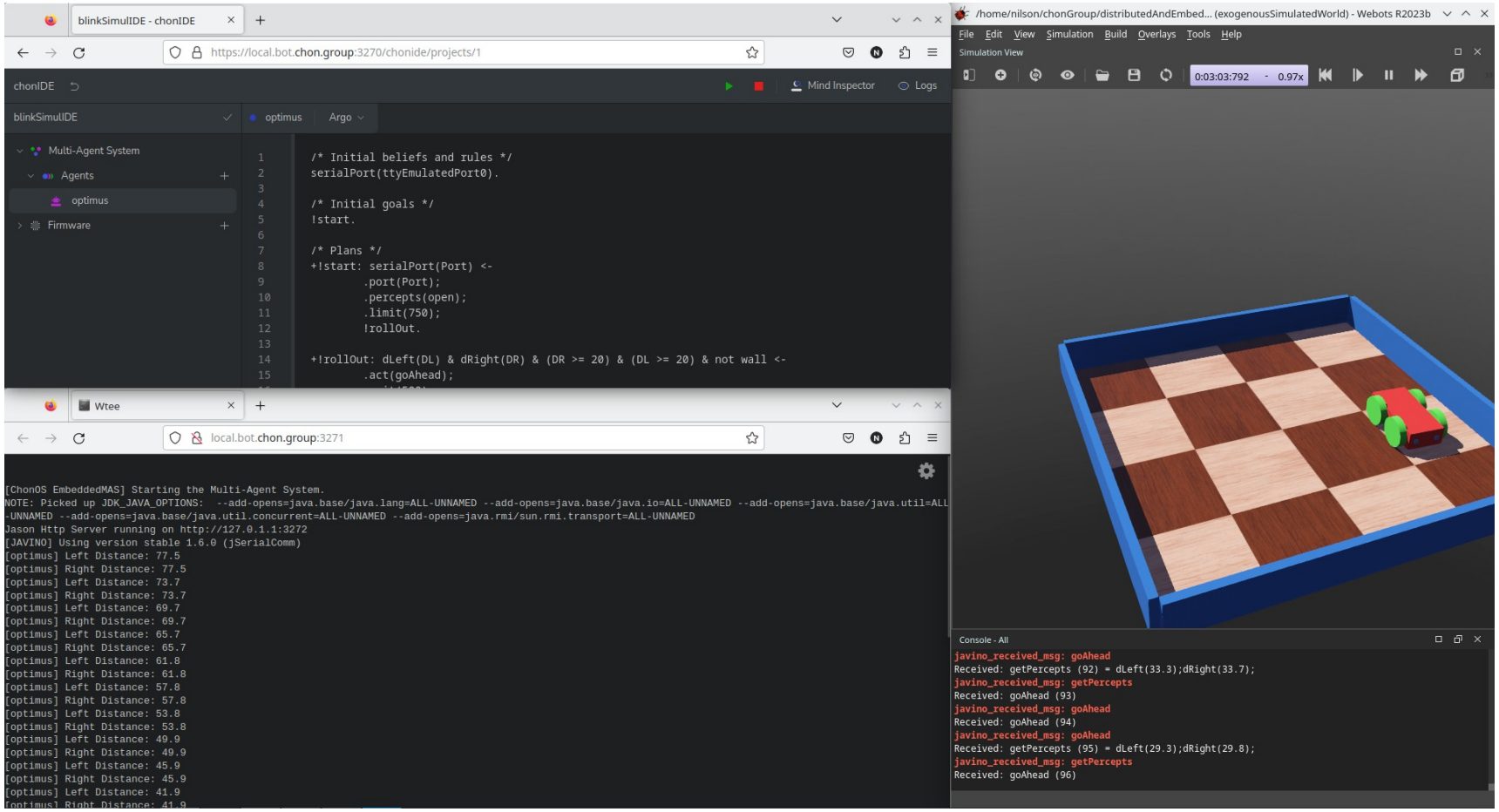


JESUS, Vinicius, Mori Lazarin, Nilson, Pantoja, Carlos Eduardo, Vaz Alves, Gleifer, Ramos Alves De Lima, Gabriel, Viterbo, Jose, An IDE to Support the Development of Embedded Multi-Agent Systems, In: Advances in Practical Applications of Agents, Multi-Agent Systems, and Cognitive Mimetics. The PAAMS Collection. Cham, 2023;13955:346–358. DOI:10.1007/978-3-031-37616-0 29.
LIMA, GABRIEL ; SIQUEIRA, ELAINE ; MEDEIROS, THÁCITO ; PANTOJA, CARLOS ; LAZARIN, NILSON ; VITERBO, JOSÉ . A Modularized and Reusable Architecture for an Embedded MAS IDE. In: 27th International Conference on Enterprise Information Systems, 2025, Porto. Proceedings of the 27th International Conference on Enterprise Information Systems, 2025. p. 1034.

The Road So Far



The Road So Far



Conclusions

- **The development of tools to support the Responsible AI/MAS engineering process**

- **Explore the Teaching and Learning Process of the Ethical Principles using BDI;**

- **These technologies have been used** in:
  - **2 undergratuate courses** (CEFET/RJ Maria da Graça and Nova Friburgo);
  - **1 graduate course** (M.Sc. and Ph.D. at UFF);
  - It was **presented at**: UTFPR, IFSP, IFC, CEFET/RJ, UFRJ, UFF, Estácio, UDESC, UFPI, and UFSC;
  - It was also **presented at**: WESAAC'22, WESAAC'23, SEKE'23, WESAAC'24, SEMESO'24, SINFO'25, WESAAC'25, and ISSIA'26.

ACKNOWLEDGMENTS

Thank You!

pantoja@cefet-rj.br

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